

Introduction

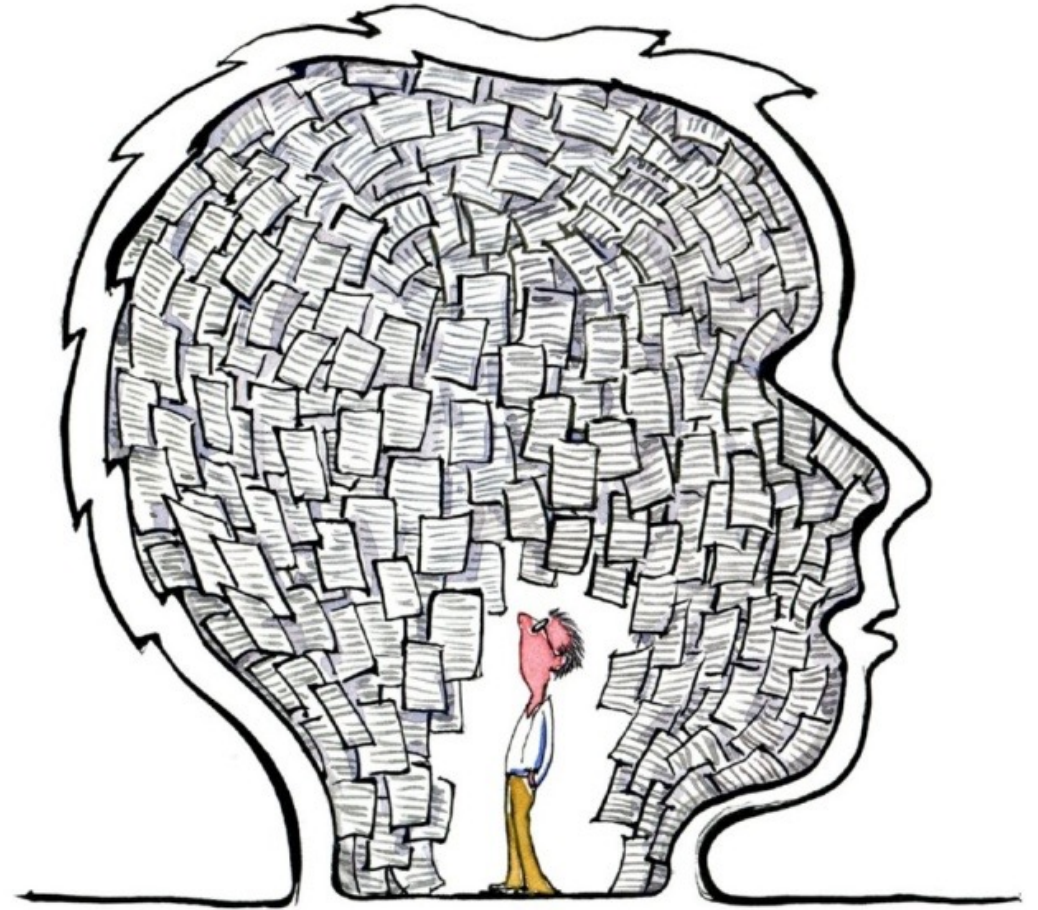
COMP7507
Visualization & Visual Analytics



SCHOOL OF
**COMPUTING &
DATA SCIENCE**
The University of Hong Kong

Data Literacy and Visualization

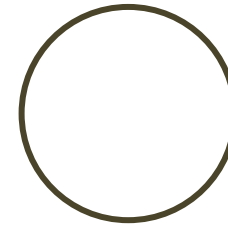
- Data literacy is the ability to read, understand, create, and communicate data as information.
- How?
- "A picture is worth a thousand words."
- Visualization is the communication of information using graphical representation.



Text vs Images

An example from Mike Parkinson's "Do-It-Yourself Billion Dollar Graphics"

*a curved line with
every point equal distance
from the center*



Text/words

- Sequential process of reading
- Limited by the use of different languages

Images/pictures

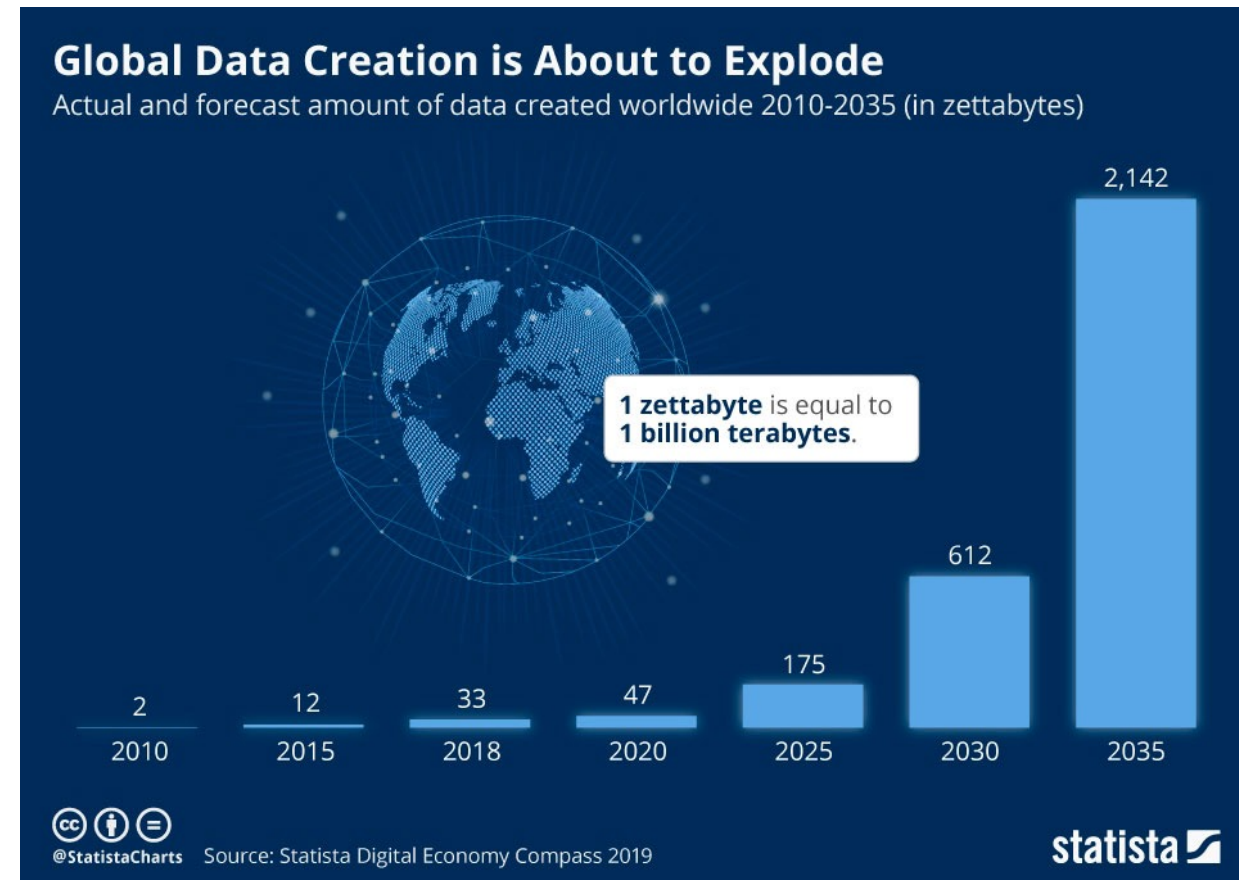
- Interpreted in parallel by the human perceptual system
- Independent of local language

The Global Datasphere

We are generating a whole lot of data every second.

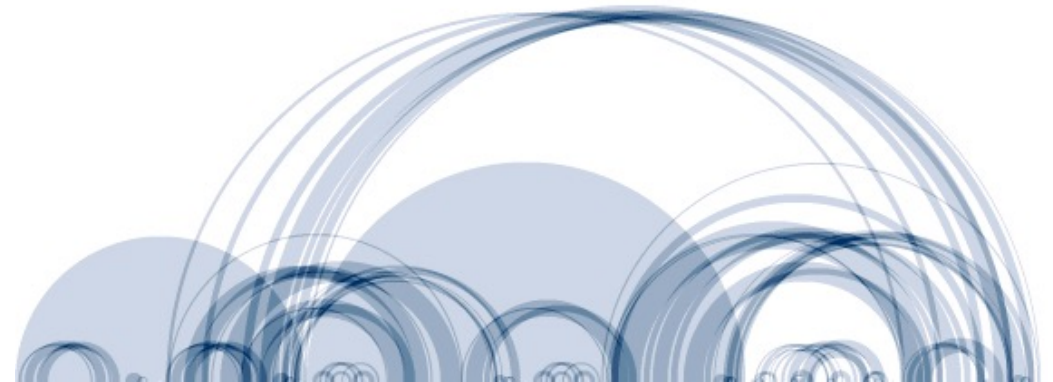
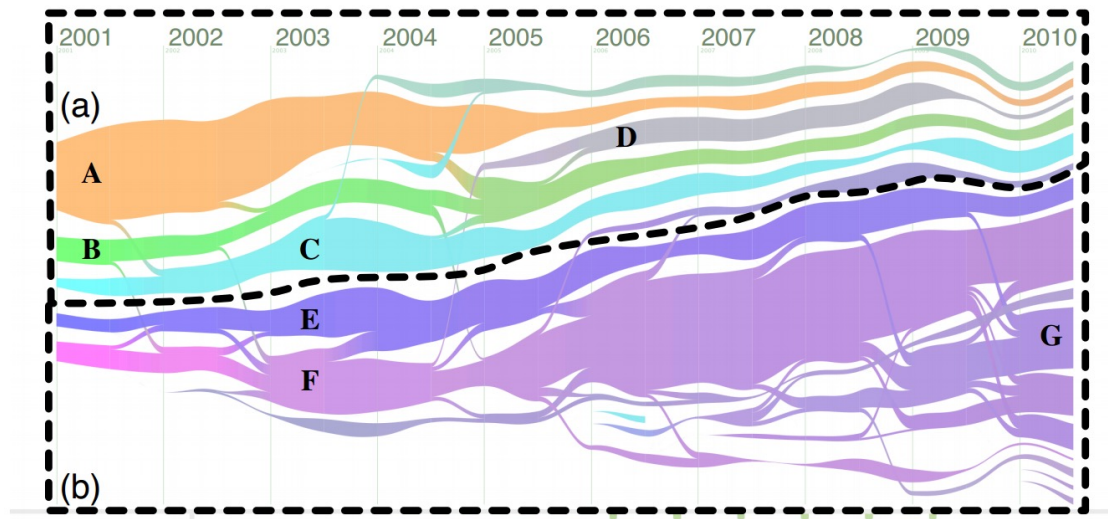
How big is the “Global Datasphere”?

- Predicted to grow from 45ZB in 2019 to 175ZB by 2025 (by IDC (International Data Corporation) in 2020)
- Predicted to grow from 149ZB in 2024 to 394ZB by 2028 (by Statista in 2024)



Data and Visualization

- Why visualization?
 - Data needs **interpretation** (e.g., to explain **process** and **concepts**)
 - Visualization serves as an effective tool to assist in **analysis** and **communication**



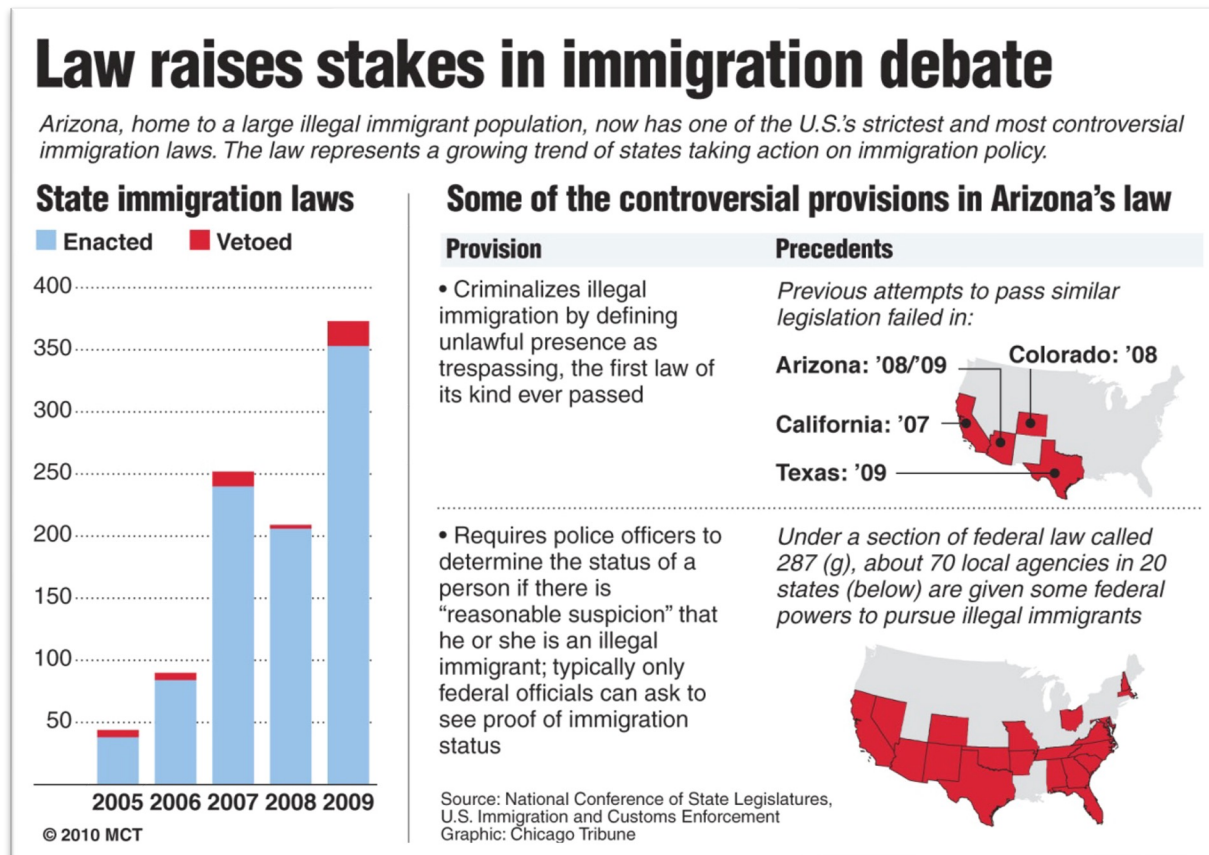
How to visualize?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	Middle "C" timing		C#		D		E ^b		E		F		F#		G	
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
3	0.014286	0.014286	0.015135	0.013873	0.016035	0.013362	0.016989	0.012741	0.017999	0.011999	0.019069	0.011123	0.020203	0.010101	0.021404	0
4	0.028571	0.028568	0.03027	0.027744	0.03207	0.026721	0.033977	0.025478	0.035998	0.023993	0.038138	0.022242	0.040406	0.020198	0.042809	0
5	0.042857	0.042844	0.045406	0.041607	0.048106	0.040072	0.050966	0.038208	0.053997	0.03698	0.057207	0.033353	0.060609	0.030286	0.064213	0
6	0.057143	0.057112	0.060541	0.055462	0.064141	0.053414	0.067955	0.050927	0.071995	0.047956	0.076277	0.044452	0.080812	0.040362	0.085618	0
7	0.071429	0.071368	0.075676	0.069303	0.080176	0.066742	0.084943	0.063631	0.089994	0.059915	0.095346	0.055534	0.101015	0.050422	0.107022	0
8	0.085714	0.085609	0.090811	0.083129	0.096211	0.080052	0.101932	0.076317	0.107993	0.071856	0.114415	0.066596	0.121218	0.060461	0.128426	0
9	0.1	0.099833	0.105946	0.096936	0.112246	0.093342	0.118921	0.08898	0.125992	0.083773	0.133484	0.077635	0.141421	0.070475	0.149831	0
10	0.114286	0.114037	0.121081	0.11072	0.128281	0.106608	0.135909	0.101619	0.143991	0.095663	0.152553	0.088645	0.161624	0.080461	0.171235	0
11	0.128571	0.128217	0.136217	0.12448	0.144317	0.119847	0.152898	0.114227	0.16199	0.107522	0.171622	0.099622	0.181827	0.090414	0.192639	0
12	0.142857	0.142372	0.151352	0.13821	0.160352	0.133055	0.169887	0.126803	0.179989	0.119346	0.190691	0.110564	0.202031	0.100329	0.214044	0
13	0.157143	0.156497	0.166487	0.151909	0.176387	0.146228	0.186875	0.139342	0.197988	0.131131	0.209761	0.121465	0.222234	0.110204	0.235448	0
14	0.171429	0.17059	0.181622	0.165573	0.192422	0.159364	0.203864	0.151841	0.215986	0.142874	0.22883	0.132322	0.242437	0.120034	0.256853	0
15	0.185714	0.184649	0.196757	0.179199	0.208457	0.172459	0.220853	0.164296	0.233985	0.154571	0.247899	0.143131	0.26264	0.129815	0.278257	0
16	0.2	0.198669	0.211893	0.192785	0.224492	0.18551	0.237841	0.176704	0.251984	0.166217	0.266968	0.153888	0.282843	0.139543	0.299661	0
17	0.214286	0.21265	0.227028	0.206326	0.240528	0.198513	0.25483	0.189061	0.269983	0.17781	0.286037	0.164589	0.303046	0.149214	0.321066	0
18	0.228571	0.226586	0.242163	0.219819	0.256563	0.211464	0.271819	0.201363	0.287982	0.189345	0.305106	0.17523	0.323249	0.158824	0.34247	0
19	0.242857	0.240477	0.257298	0.233263	0.272598	0.224362	0.288807	0.213607	0.305981	0.200819	0.324175	0.185808	0.343452	0.16837	0.363875	0
20	0.257143	0.254318	0.272433	0.246653	0.288633	0.237202	0.305796	0.225789	0.32398	0.212228	0.343245	0.196317	0.363655	0.177846	0.385279	0
21	0.271429	0.268108	0.287569	0.259986	0.304668	0.249981	0.322785	0.237907	0.341979	0.223568	0.362314	0.206756	0.383858	0.18725	0.406683	0
22	0.285714	0.281843	0.302704	0.27326	0.320703	0.262695	0.339773	0.249955	0.359977	0.234835	0.381383	0.217119	0.404061	0.196578	0.428088	0
23	0.3	0.29552	0.317839	0.286472	0.336739	0.275342	0.356762	0.261932	0.377976	0.246027	0.400452	0.227404	0.424264	0.205825	0.449492	0
24	0.314286	0.309137	0.332974	0.299617	0.352774	0.287918	0.373751	0.273832	0.395975	0.257139	0.419521	0.237605	0.444467	0.214988	0.470897	0
25	0.328571	0.322691	0.348109	0.312694	0.368809	0.300421	0.390739	0.285654	0.413974	0.268167	0.43859	0.24772	0.46467	0.224064	0.492301	0
26	0.342857	0.336179	0.363244	0.3257	0.384844	0.312846	0.407728	0.297394	0.431973	0.279109	0.457659	0.257746	0.484873	0.233048	0.513705	0
27	0.357143	0.349599	0.37838	0.338631	0.400879	0.32519	0.424717	0.309047	0.449972	0.28996	0.476729	0.267677	0.505076	0.241937	0.53511	0
28	0.371429	0.362947	0.393515	0.351484	0.416914	0.337451	0.441705	0.320611	0.467971	0.300717	0.495798	0.277511	0.525279	0.250727	0.556514	0
29	0.385714	0.376221	0.40865	0.364257	0.43295	0.349625	0.458694	0.332083	0.48597	0.311377	0.514867	0.287244	0.545482	0.259415	0.577918	0
30	0.4	0.389418	0.423785	0.376946	0.448985	0.361709	0.475683	0.343459	0.503968	0.321936	0.533936	0.296873	0.565685	0.267997	0.599323	0

- Data in tabular form is hard to discover relationship, trends, patterns

Daily Life Examples of Visualization

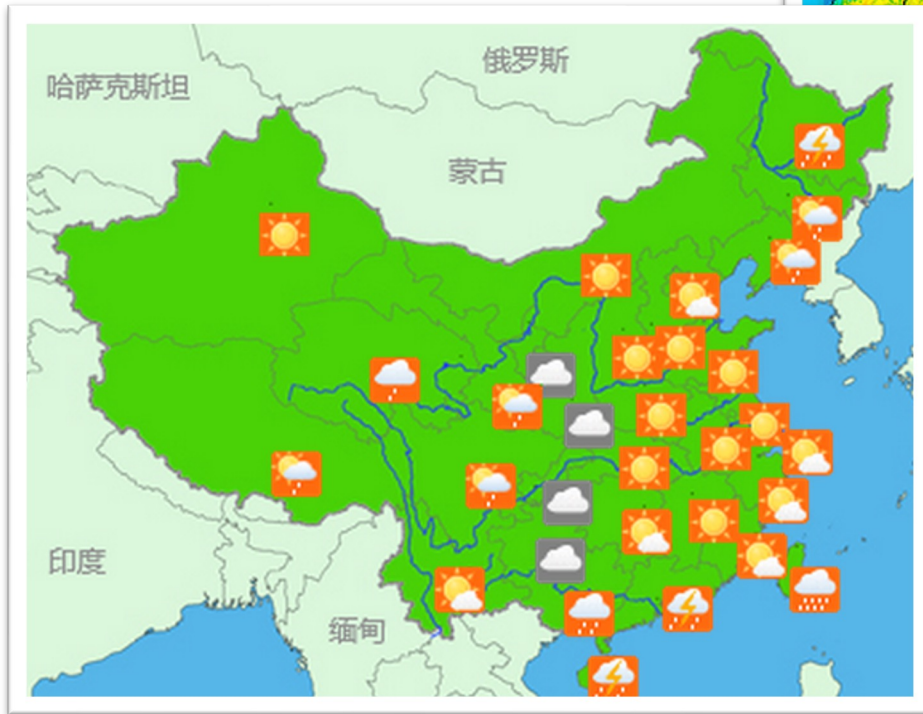
- Newspaper / Magazine – in articles accompanying text to present data



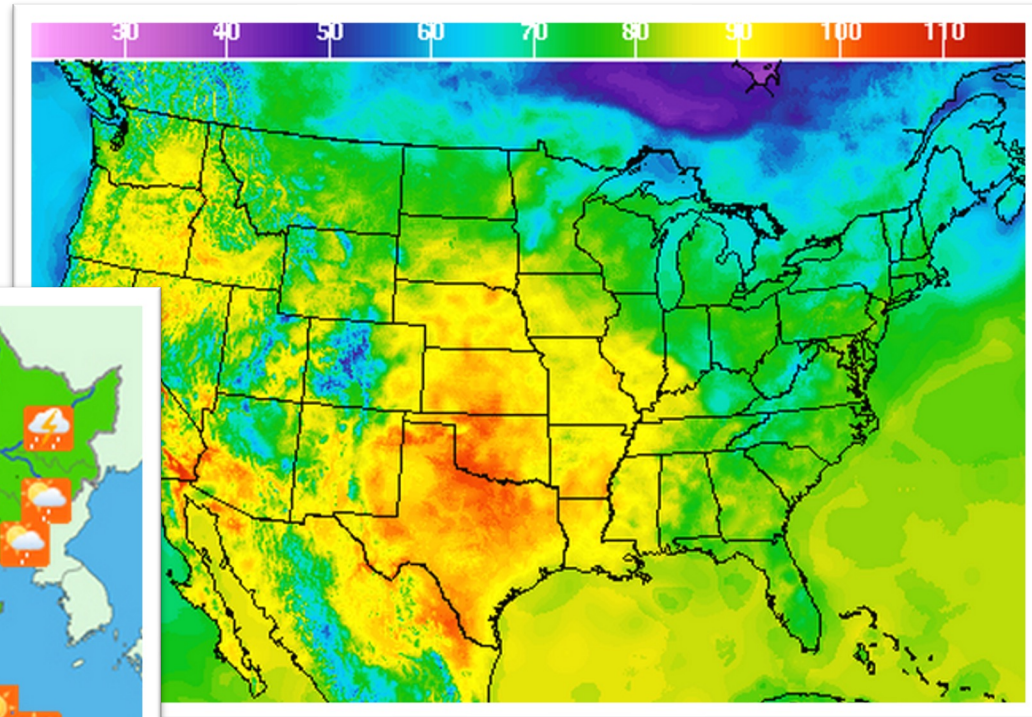
Simple & static

Daily Life Examples of Visualization

- Weather chart



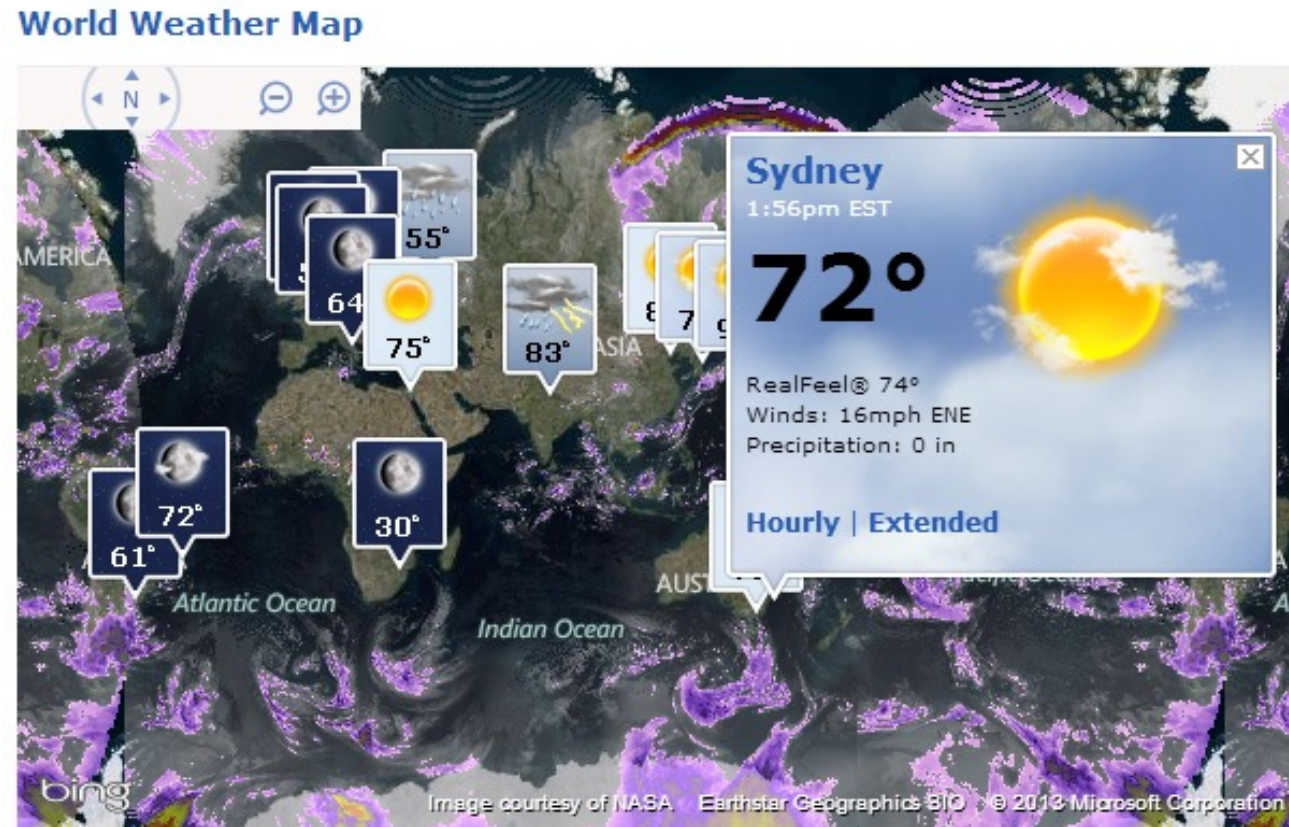
High level abstraction



Low level specifics

Keep **the user** in mind.
What do they want?

Daily Life Examples of Visualization



Occlusion problems when too much information needs to be shown.
Use **interaction** to resolve.

Daily Life Examples of Visualization

- Stock market

Complex presentation

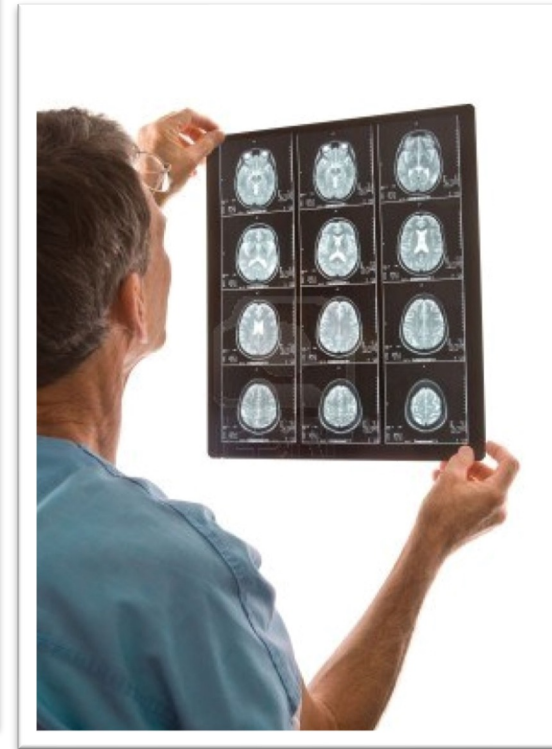
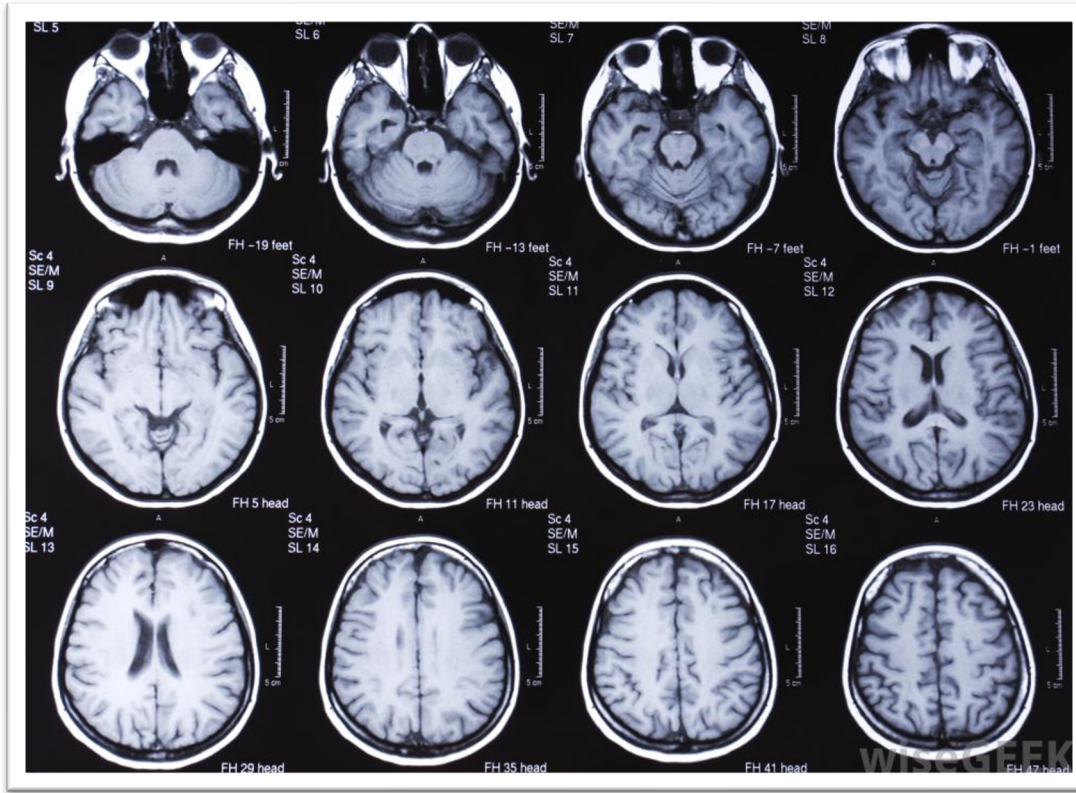
Designed for target
users with proper
training

Every visualization has
a **purpose**



Daily Life Examples of Visualization

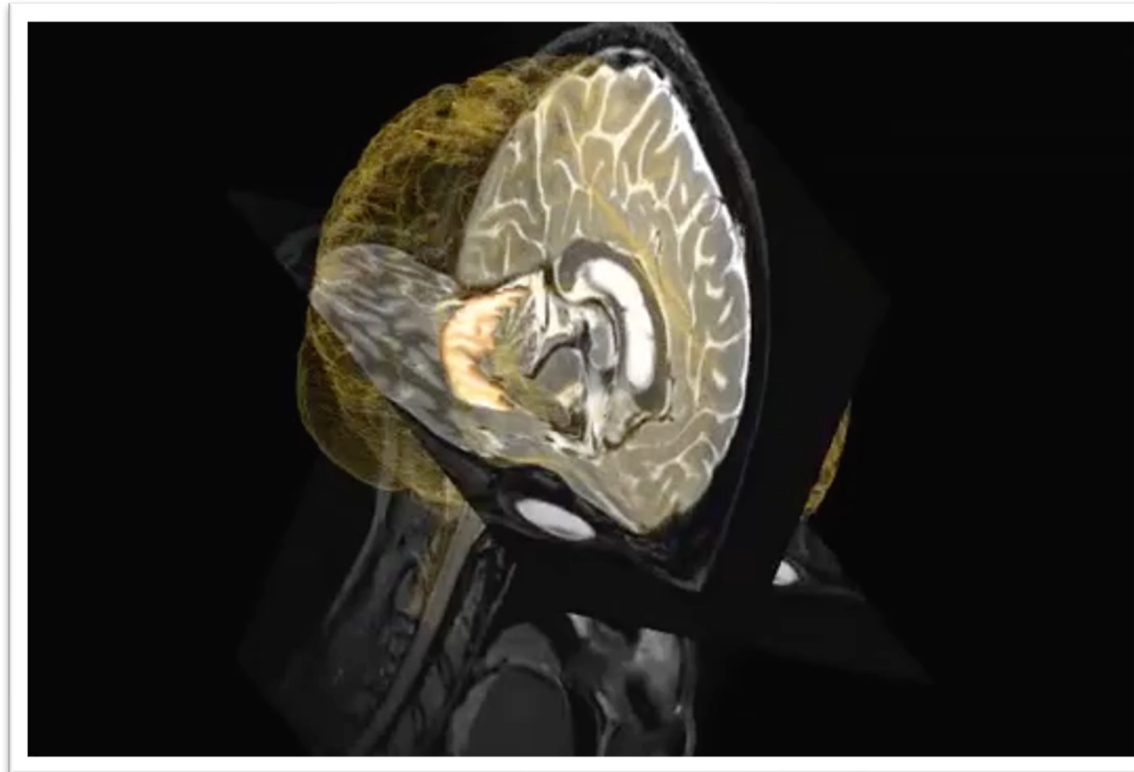
- Medical Care (Brain MRI 2D)



Need expertise and training for spatial association

Daily Life Examples of Visualization

- Medical care (Brain MRI 3D Visualization)



[VUmc Amsterdam, <http://www.youtube.com/watch?v=wjvDDH-uJ0s>]

3D graphics
embedded with the
actual data

Changing viewpoint
and object
transparency helps
show the

context

Daily Life Examples of Visualization

- Medical care (Ultrasound)

2D



https://www.youtube.com/watch?v=8if_IVf0yHw

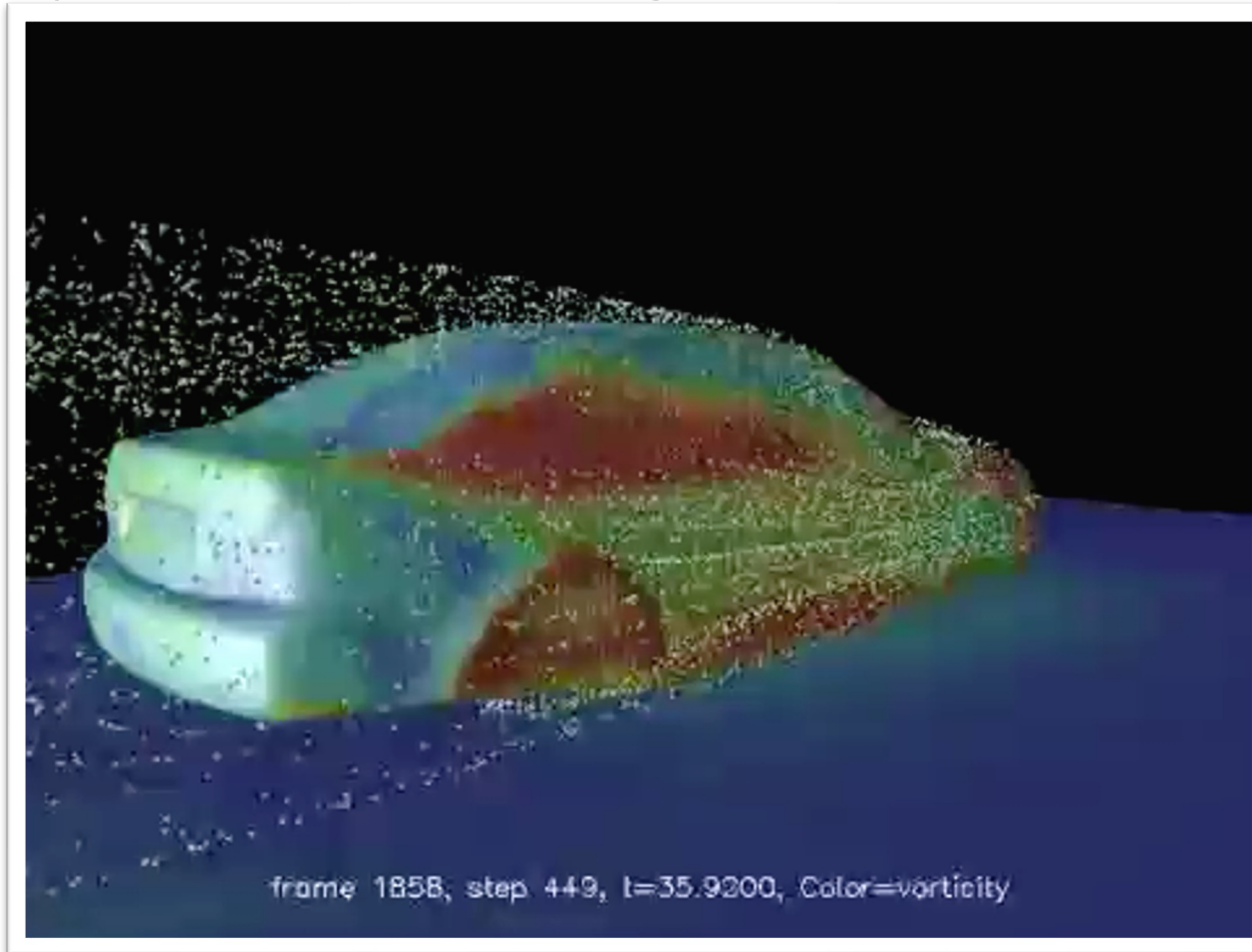
3D/4D



https://www.youtube.com/watch?v=nD_I0EtKtNYU

Examples of Visualization at Work

- Aerodynamics of a racing car

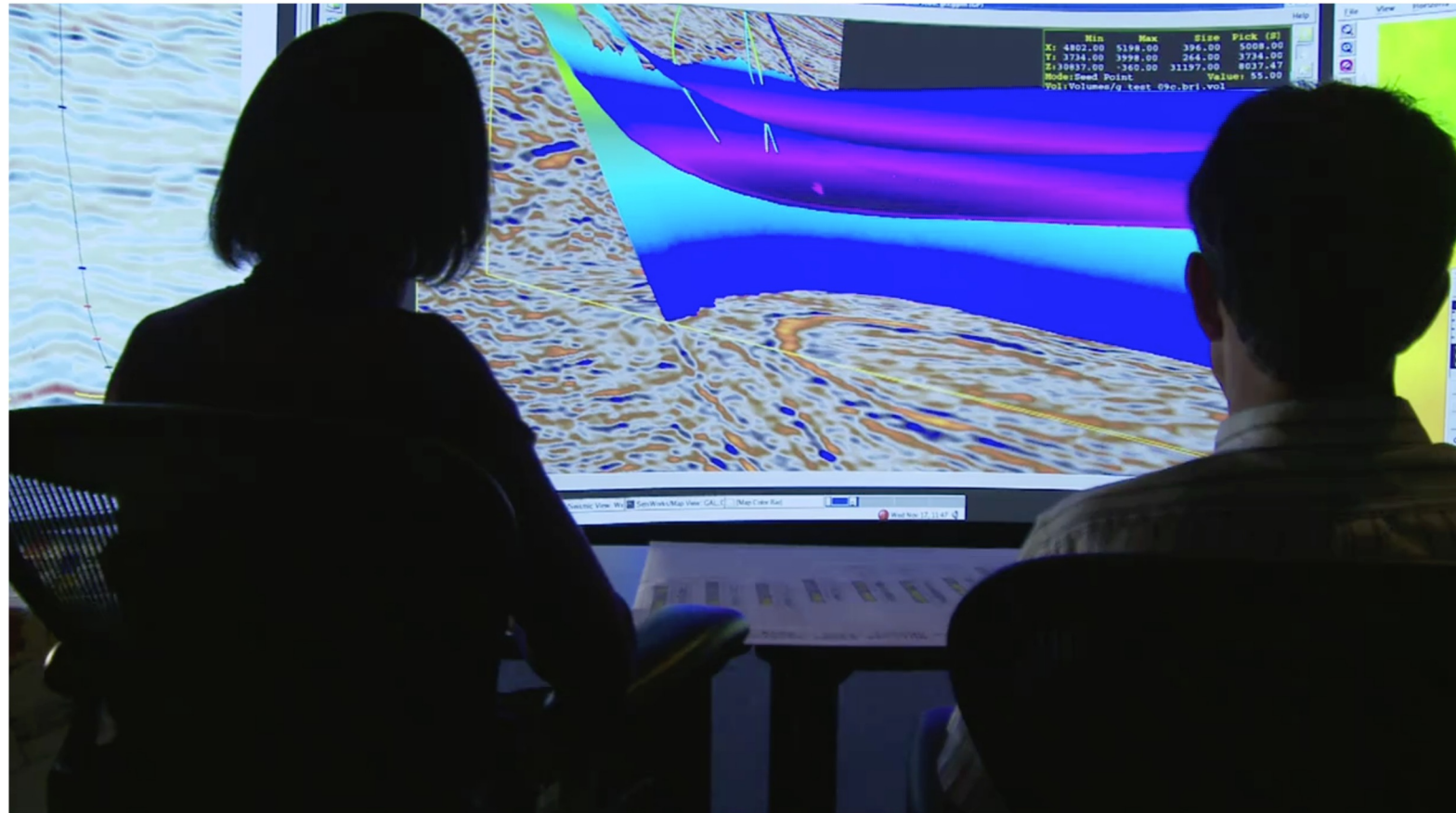


Color encodes physical properties/quantities, e.g., vorticity

Visual assistance
for checking if the result is as expected

Examples of Visualization at Work

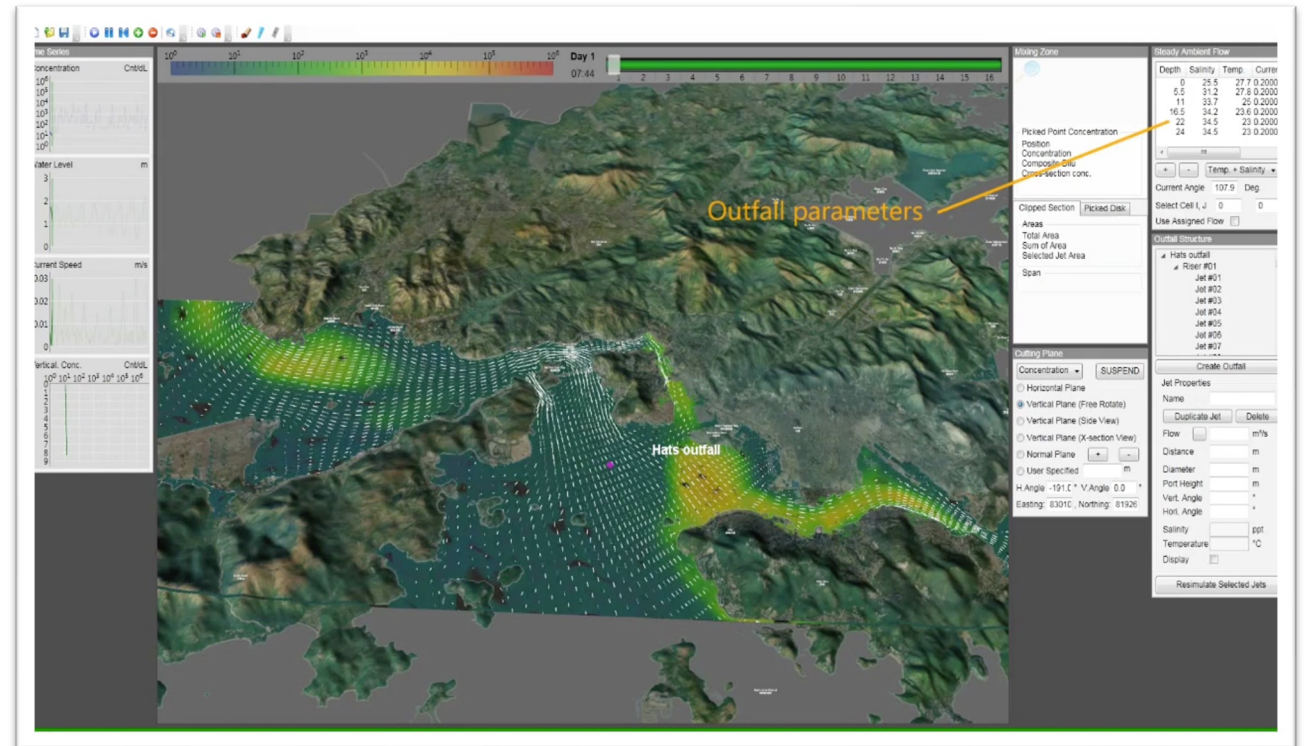
- Geophysicist in hunt for oil and gas



Immersive system
with big displays for
exploratory visualization

An Example at HKU CS

- Project WATERMAN
(www.waterman.hku.hk)
 - An Internet and GIS-based water quality forecast and management system for HK
 - Uses intensive visualization techniques for 3D Environmental Impact Assessment



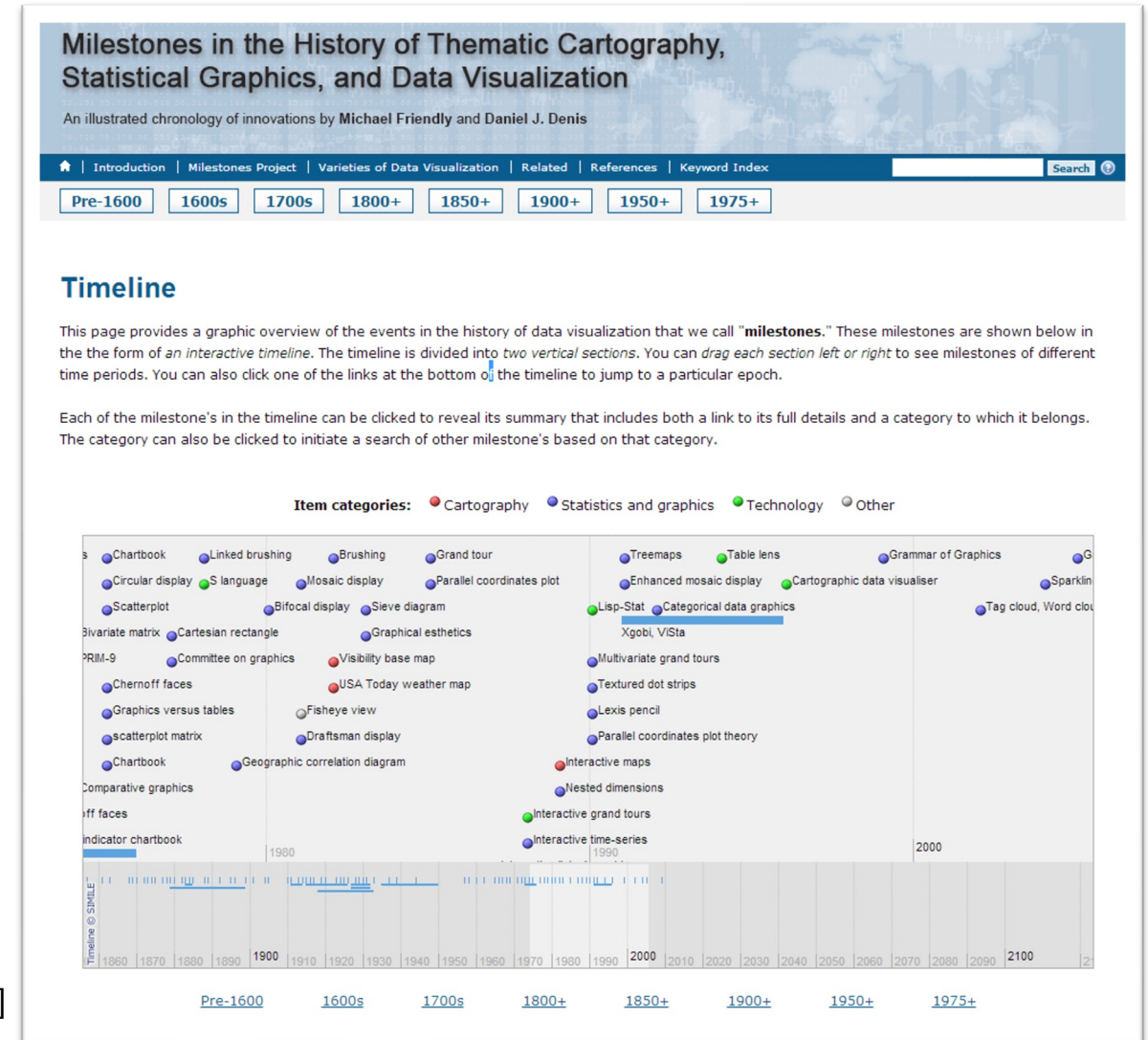
Some Questions

- Why can visualization explain better than text?
- Is visualization always effective?
- Given the same data, are there different ways to visualize it?
Which is the best?

History of Visualization

- Milestones in the history of thematic cartography, statistical graphics, and data visualization
- From 6200 BC to present

[<http://www.datavis.ca/milestones/>]



The French Invasion of Russia in 1812

- French invaded Russia in June 1812, size of the Grand Army = 422K, starting from Polish-Russian border
- The Napoleon army (size = 100K) reached Moscow in Sept
- The army retreated from Moscow in Oct, and underwent severely cold weather and lack of supplies
- The army crossed the Berezina River in Nov with only 27K soldiers left
- Finally back to Poland with only 10K men remaining



A **story** to be told

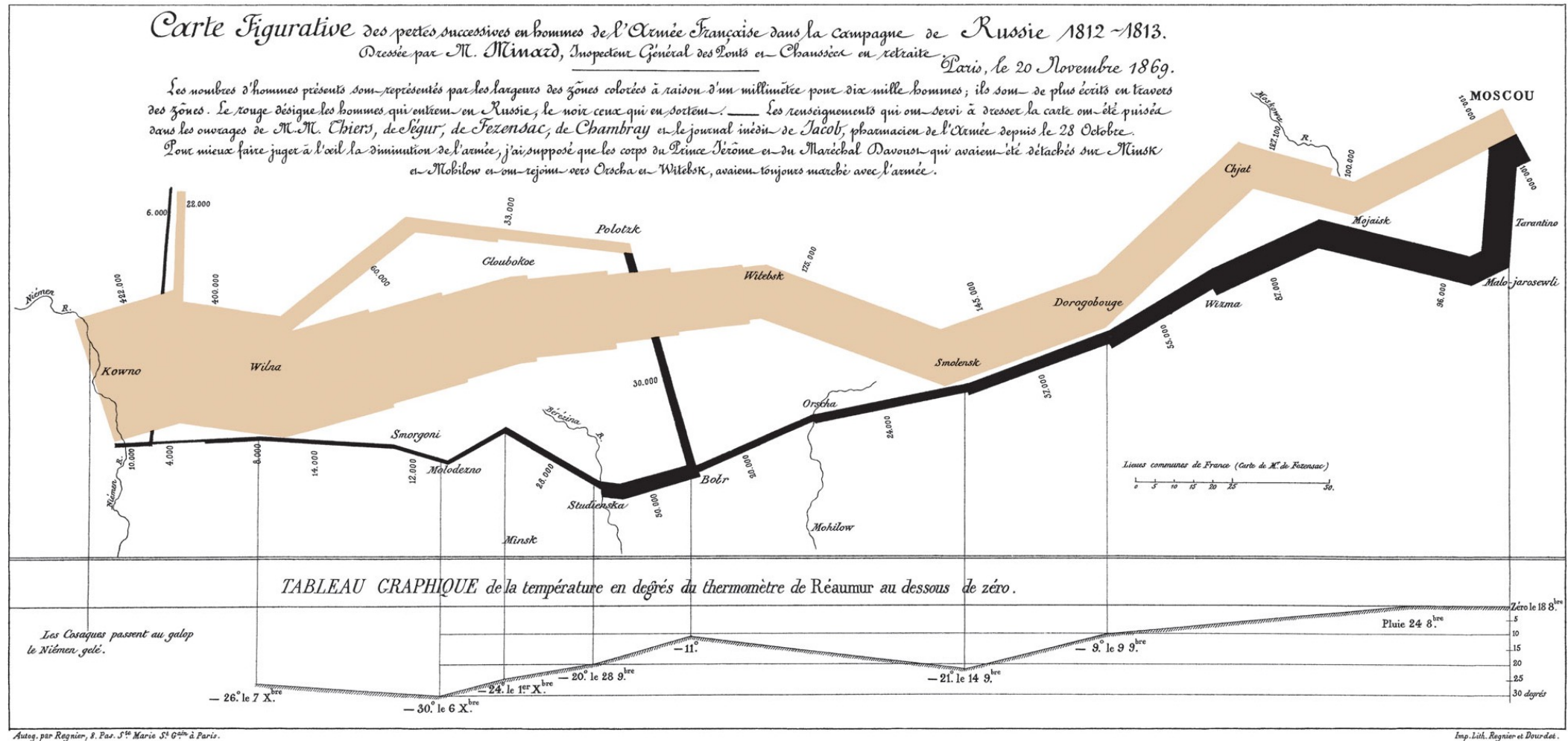
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- Variables in the story
 - Timeline of the story
 - Size of the army
 - Location of the army (2D)
 - Direction of the army's movement (2D)
 - Temperature during the retreat from Moscow
- How to present in one single drawing?

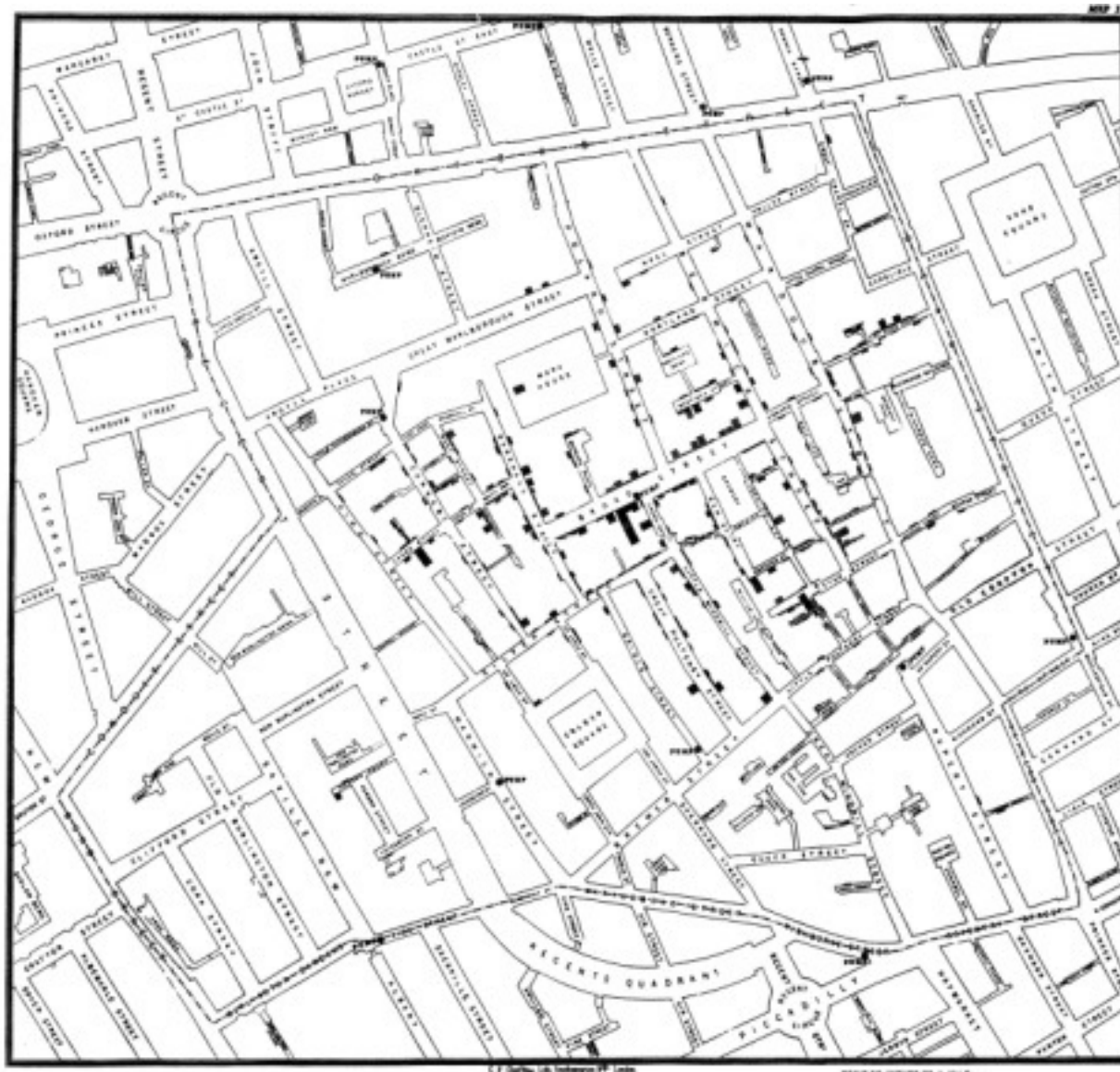
The French Invasion of Russia in 1812

- The classic graphic by Charles Joseph Minard (1781-1870)



The Cholera Epidemic in London, 1854

- Dr. John Snow's plot of over 500 deaths in central London from cholera (霍亂) in Sept 1854.
- The plot helped identifying the concentration of deaths near the Broad Street water pump.



Original map by Snow



Pump locations
marked as crosses

Concentration of death
leads to the hypothesis
that the disease can be
spread by water

Outliers can be
identified easily for
further **verification of
the hypothesis**

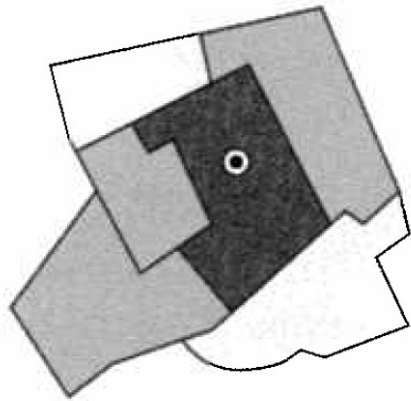
A simplified version by E.W. Gilbert

The Cholera Epidemic in London, 1854

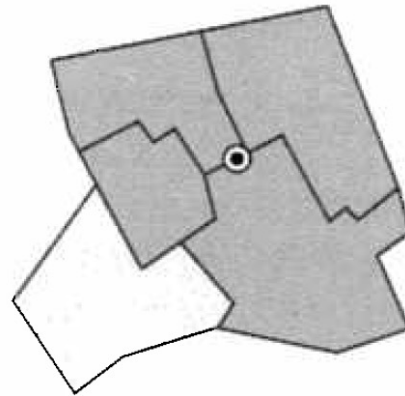
- Snow used the dot plot, marking each individual death.
 - Dots are cluttered
 - Death rates are not shown
- Question: If a region is without dots, does it mean that there is no deaths or that the area is not populated?
 - Solution: To show the death rates in a region
 - However, this requires the **aggregation of data**, e.g., to sum up the population within a certain area.

The Cholera Epidemic in London, 1854

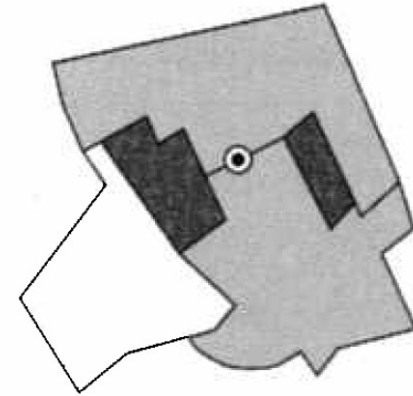
- **Aggregation** may sometimes mask relevant details, distort the actual data, and generate misleading signals.



In this aggregation of individual deaths into six areas, the greatest number is concentrated at the Broad Street pump.



Using different geographic subdivisions, the cholera numbers are nearly the same in four of the five areas.



In this aggregation of the deaths, the two areas with the most deaths do not even include the infected pump!

¹⁸ Mark Monmonier, *How to Lie with Maps* (Chicago, 1991), pp. 142–143.

Why Visualization is Important?

- provides an ability to **comprehend** huge amounts of data
- allows the perception of emergent properties that were not anticipated (**new insight**)
- often reveals problems with the data itself quickly (**anomalies**)
- facilitates **understanding** of **both large-scale and small-scale features** of the data
- facilitates **hypothesis formation**

What Should a Good Visualization Achieve?

- **Show** the data
- Induce the viewer to focus on the **substance** rather than the methodology, graphic design, the technology of graphic production, etc.
- **Avoid distorting** the data
- Present many numbers in a small space
- Encourage the eye to **compare** different pieces of data
- Reveal the data at several **levels of detail**, from a broad overview to the fine structure
- Serve a reasonably **clear purpose**: description, exploration, tabulation, or decoration

Some Bad Examples

- Data:
 - the percentage of college students in US that were under 25 from 1972 through 1976

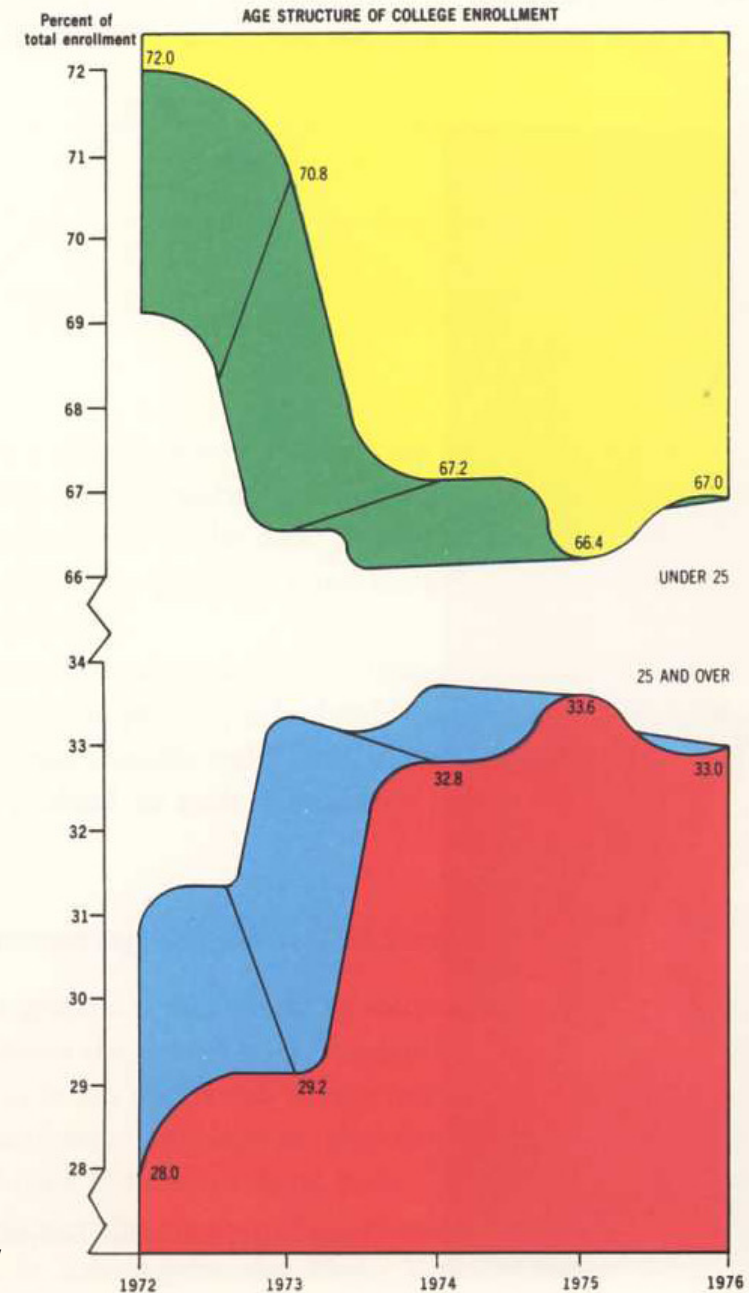
Year	Percentage
1972	72.0
1973	70.8
1974	67.2
1975	66.4
1976	67.0

Some Bad Examples

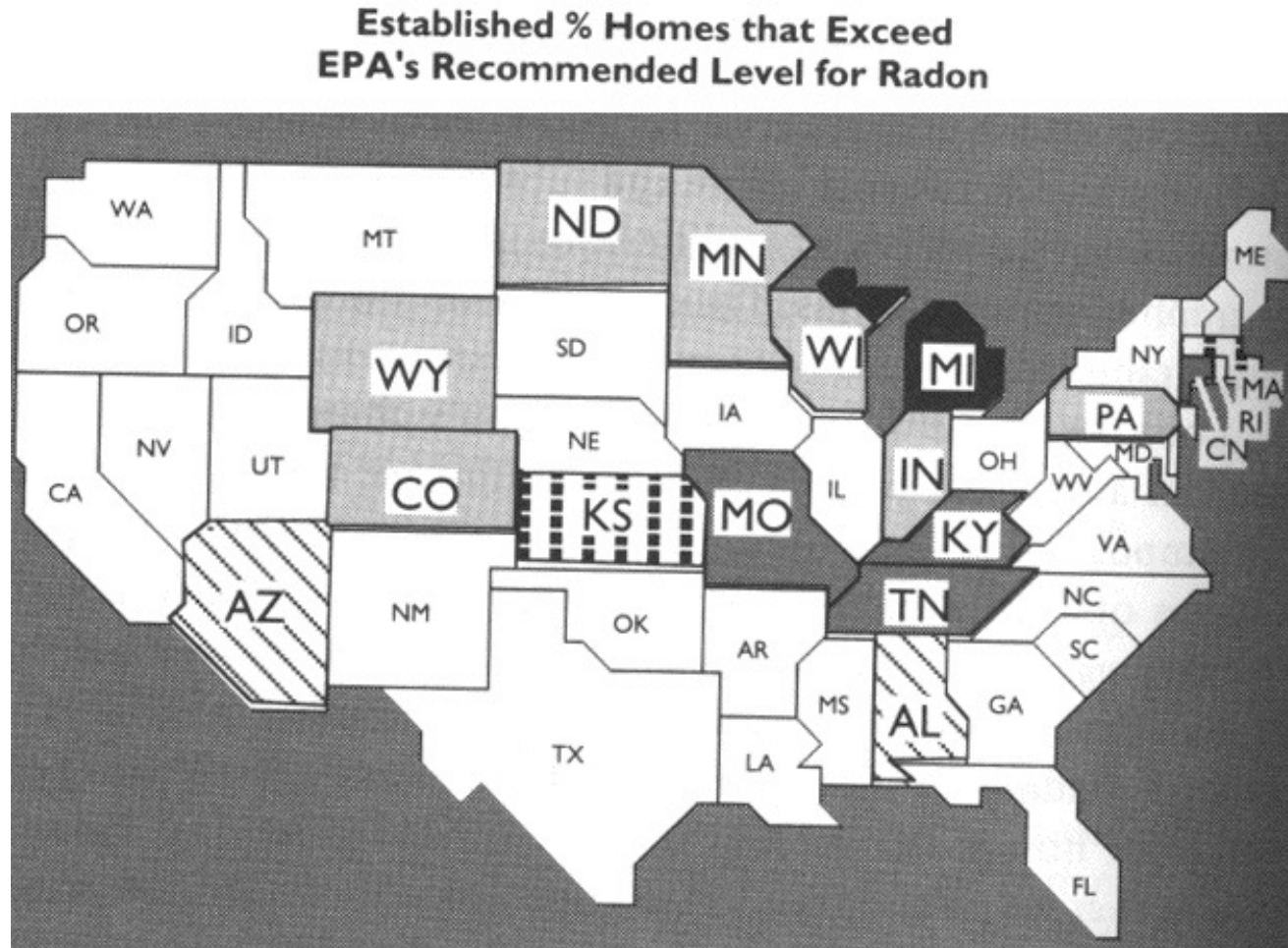
Simple guideline:

- Use table to show small data set with < 20 numbers
- Use graphic for large data set

Tufte (1983, p.118) says, "This may be the worst graphic ever to find its way into print."



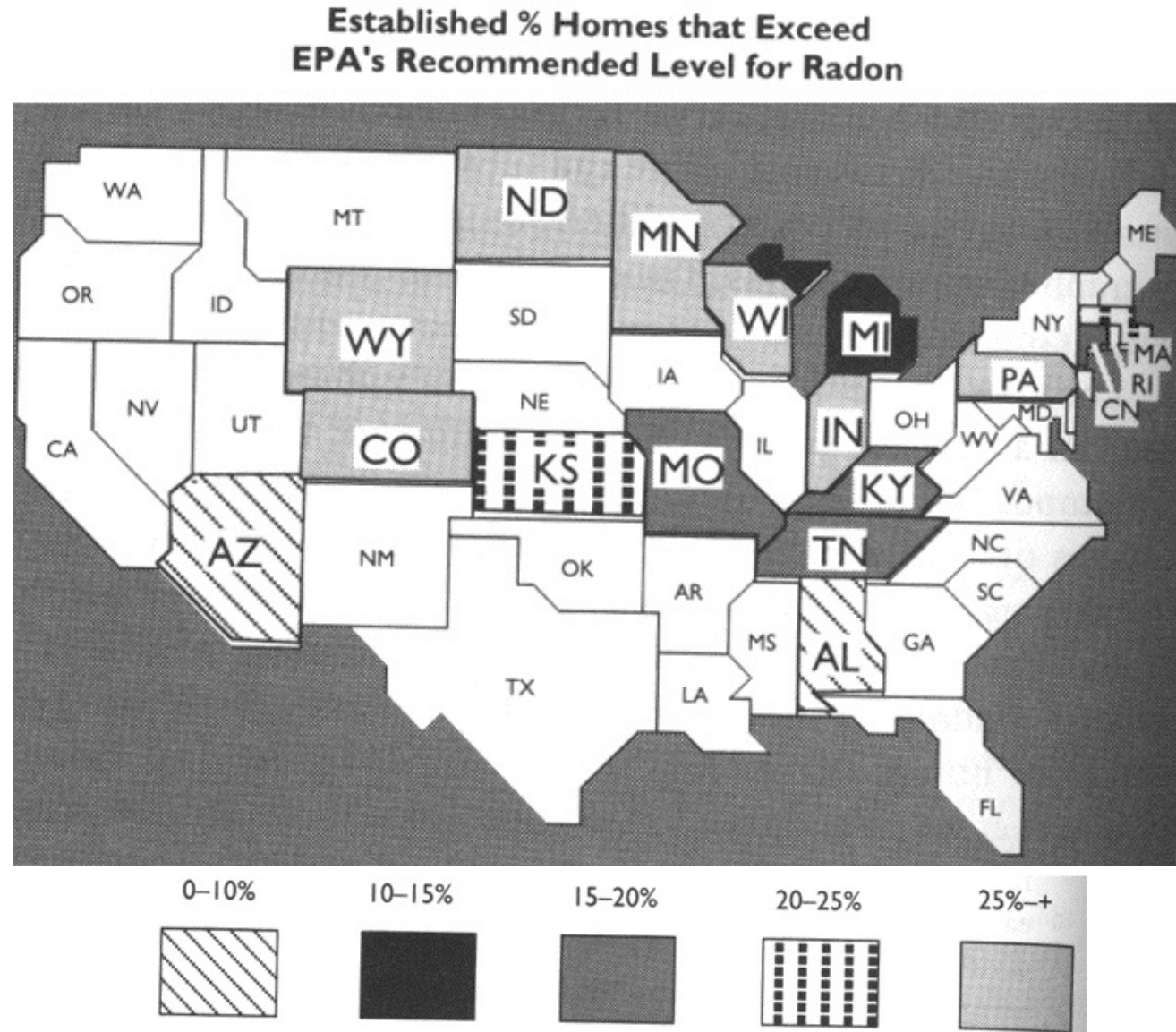
Some Bad Examples



Which state in US has the highest % of homes exceeding the recommended Radon level? Which one has the lowest?

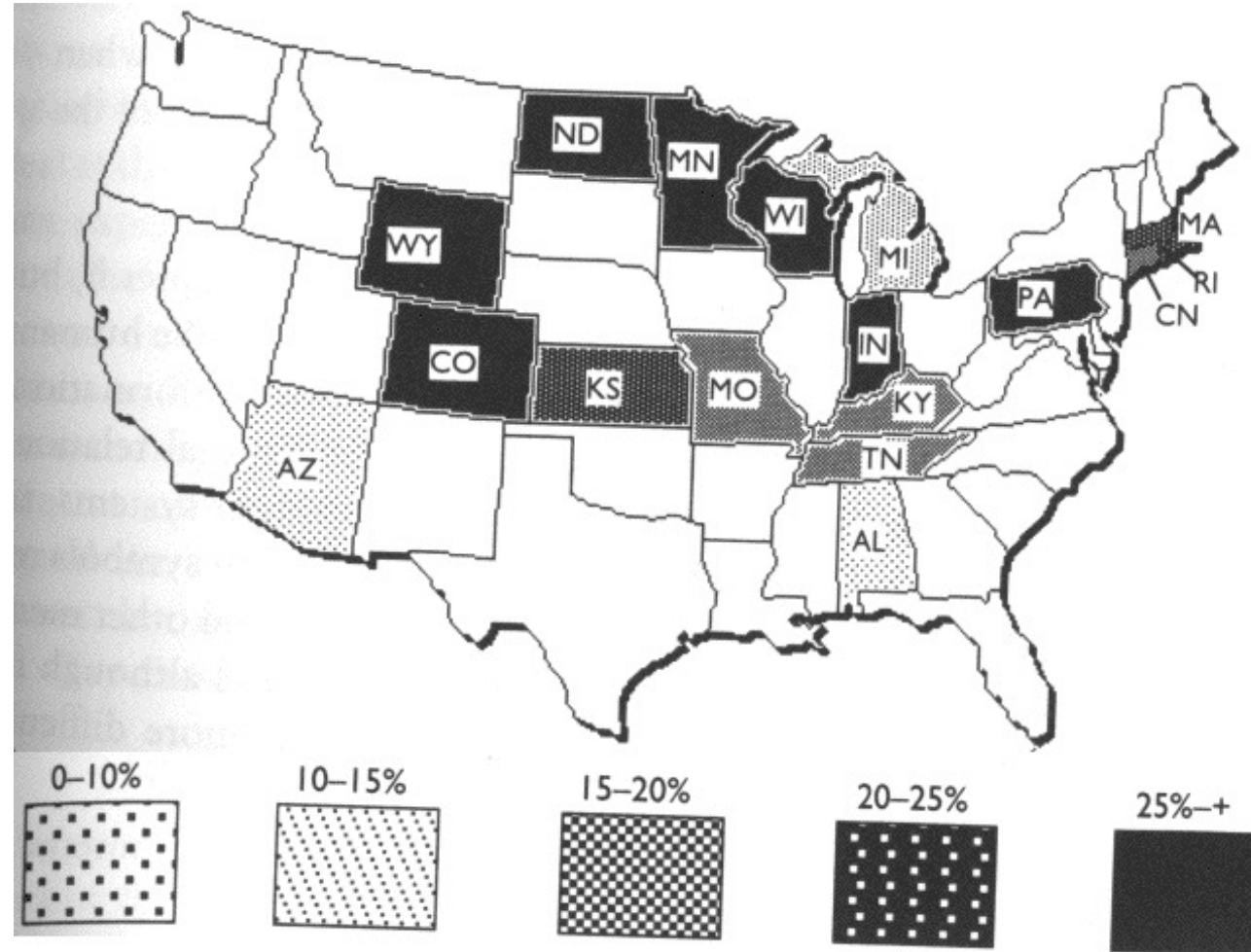
[Things that Make Us Smart, p70-71]

Some Bad Examples



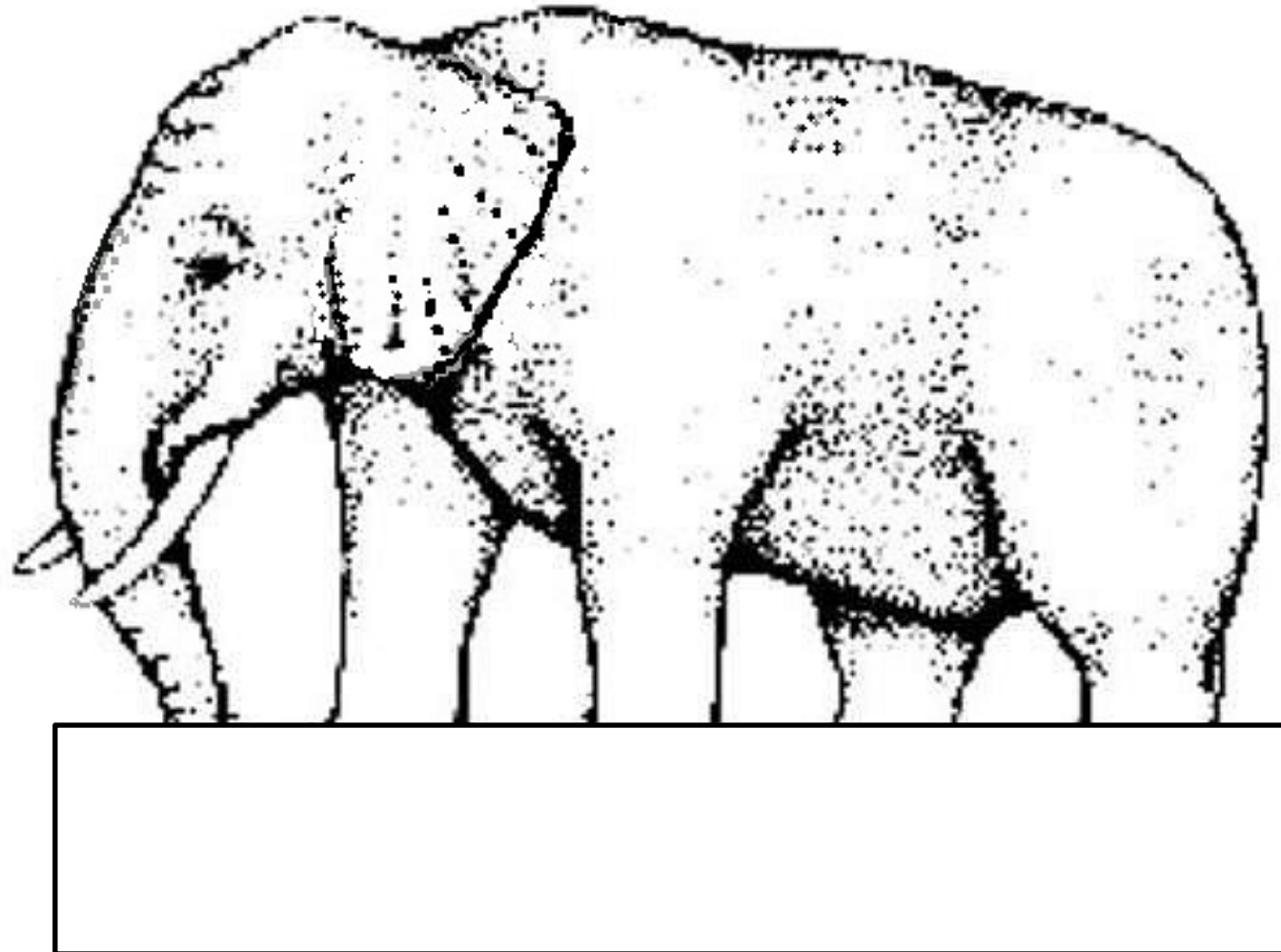
Some Bad Examples

- A much better presentation

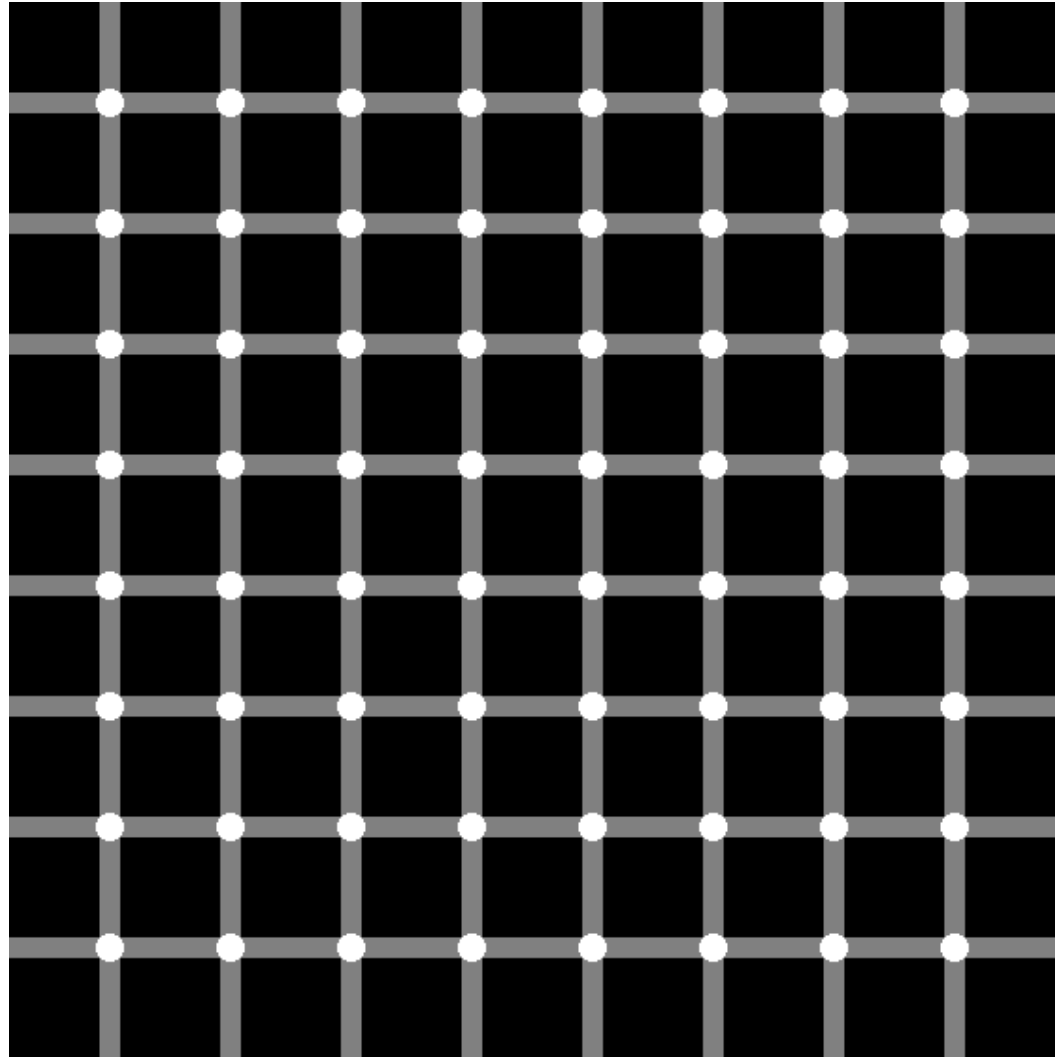


Graphics Illusion

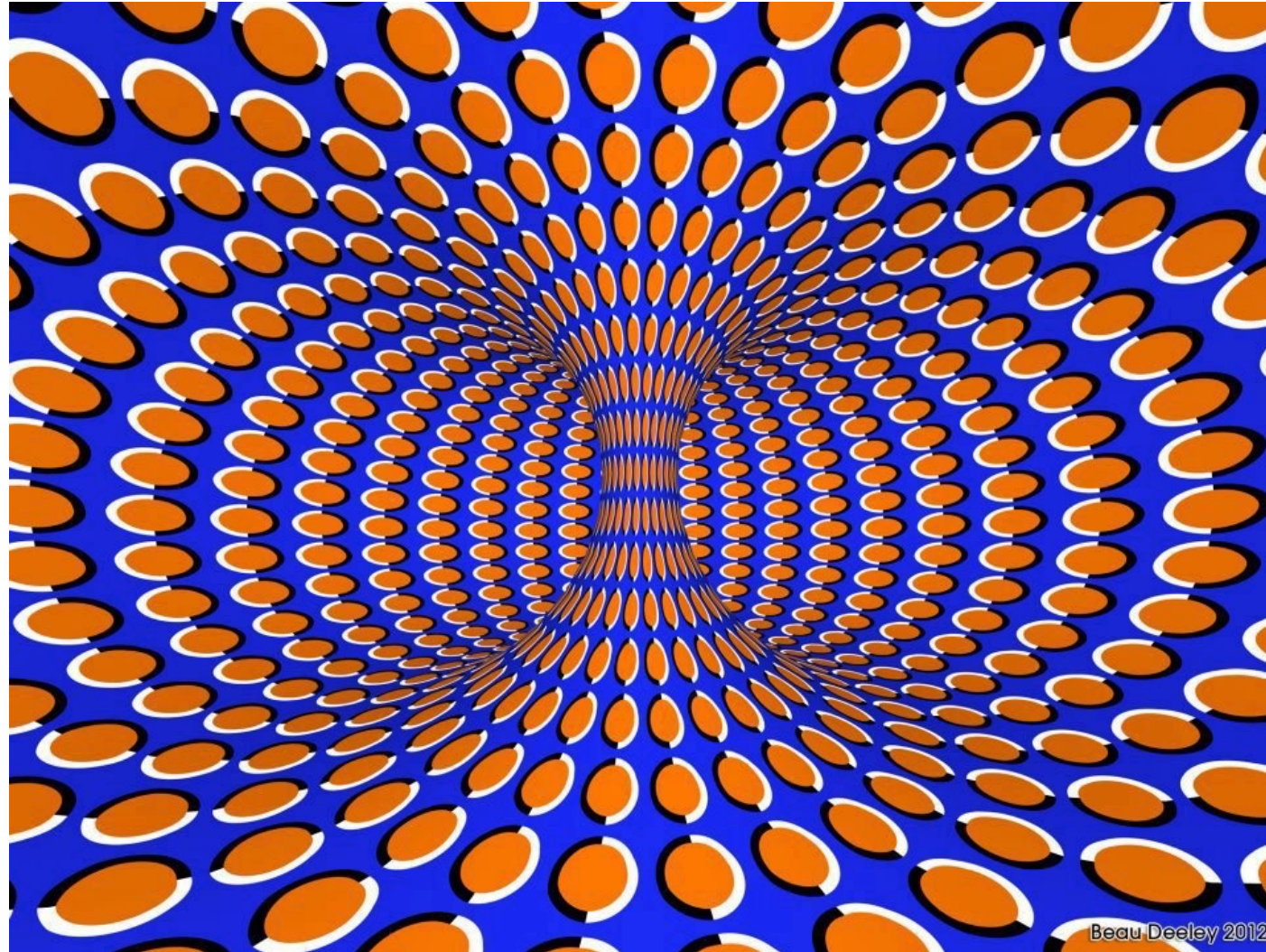
- Local vs. Global



Graphics Illusion

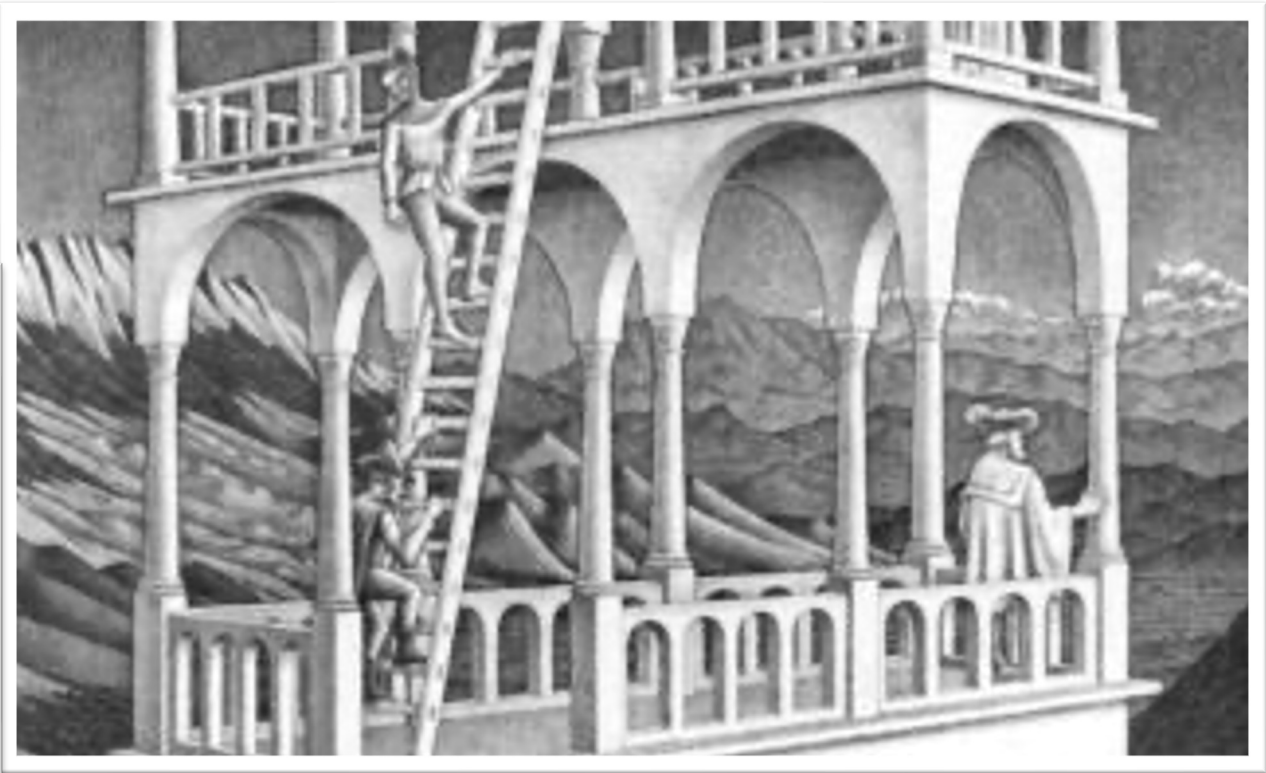
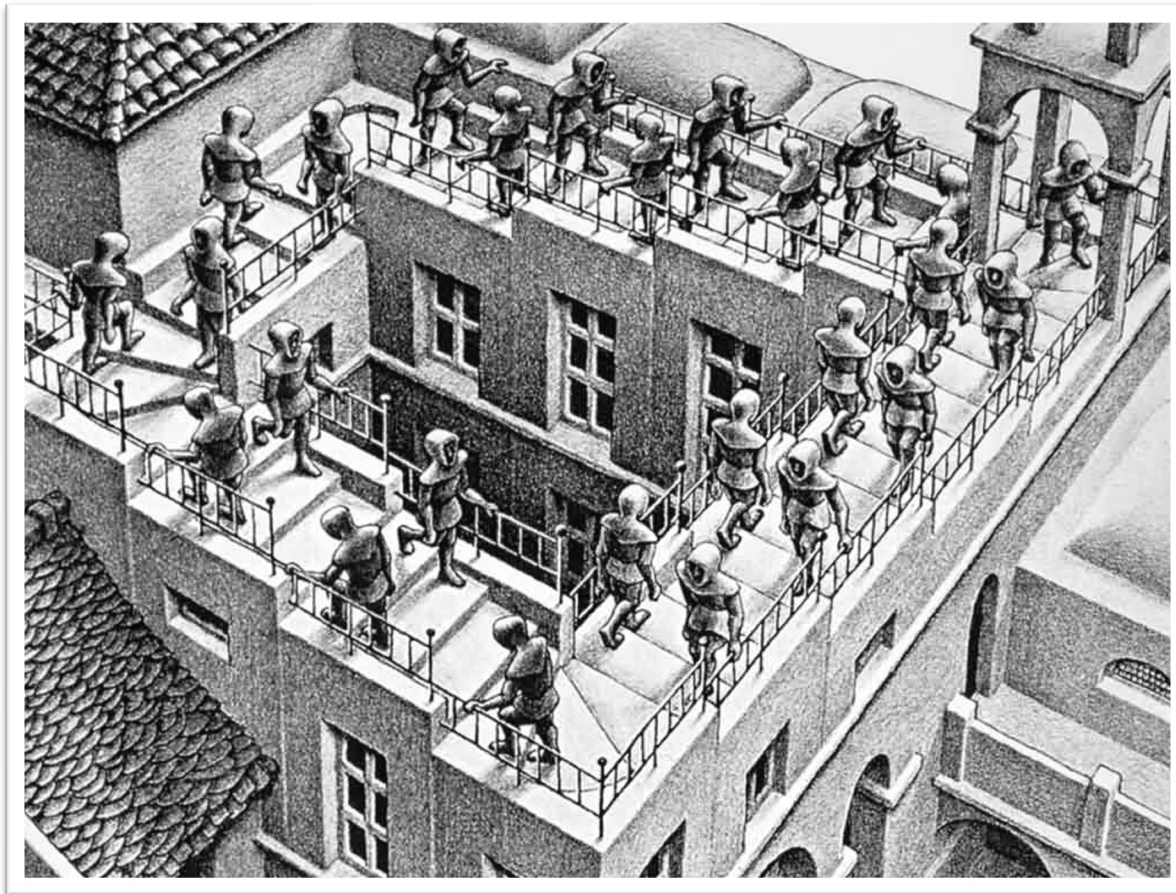


Graphics Illusion



Beau Deeley 2012

M.C. Escher (1898-1972)



Escher for Real

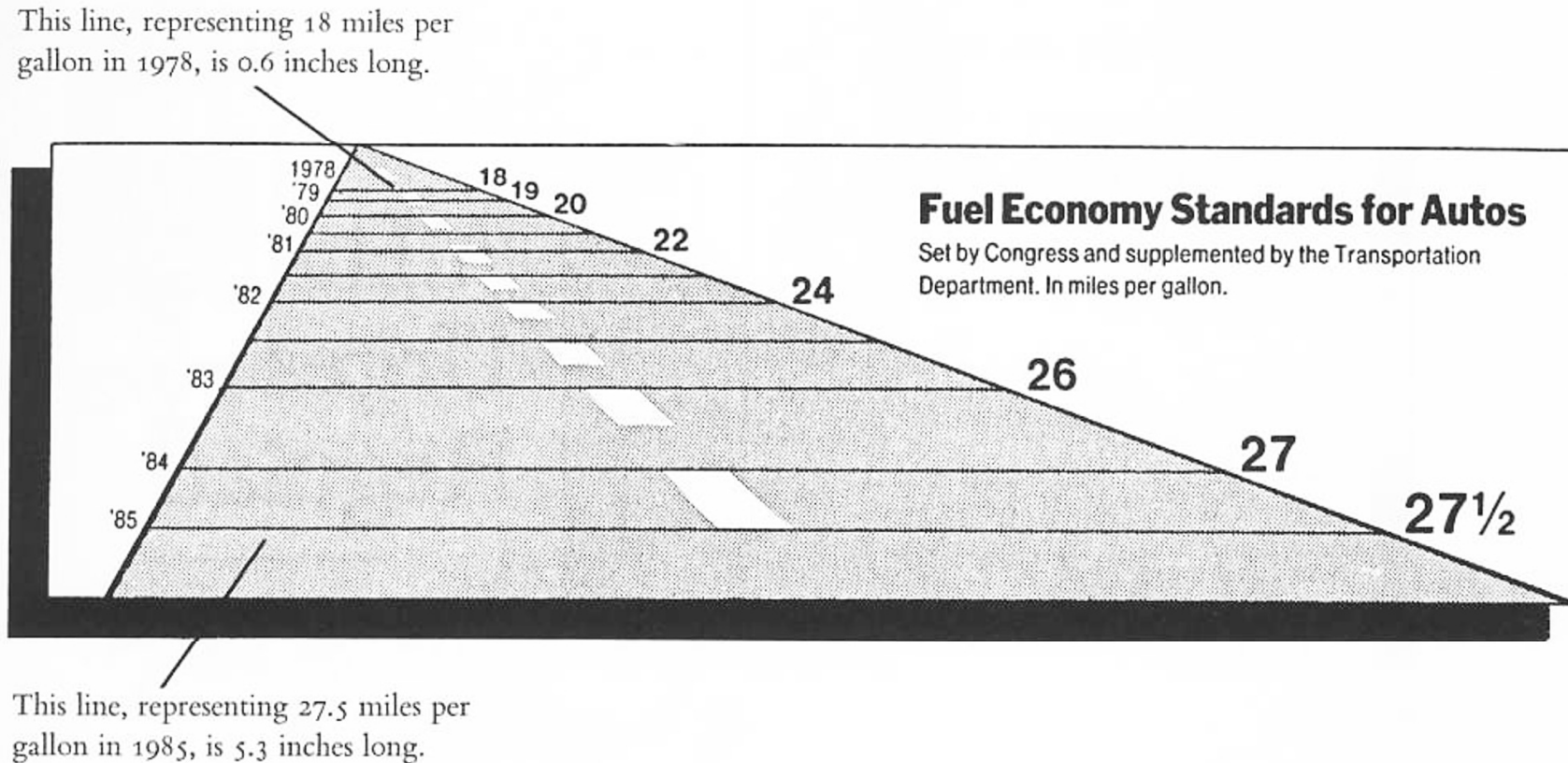


<http://gershoner.org/EscherForReal/>

Graphic that fails to tell the truth

- A graphic does not distort if the visual representation of the data is **consistent** with the numerical representation.
- What is visual representation?
 - Physically measured?
 - Perceived visual effect?
- One thing to note: Perception changes with experience and is context-dependent

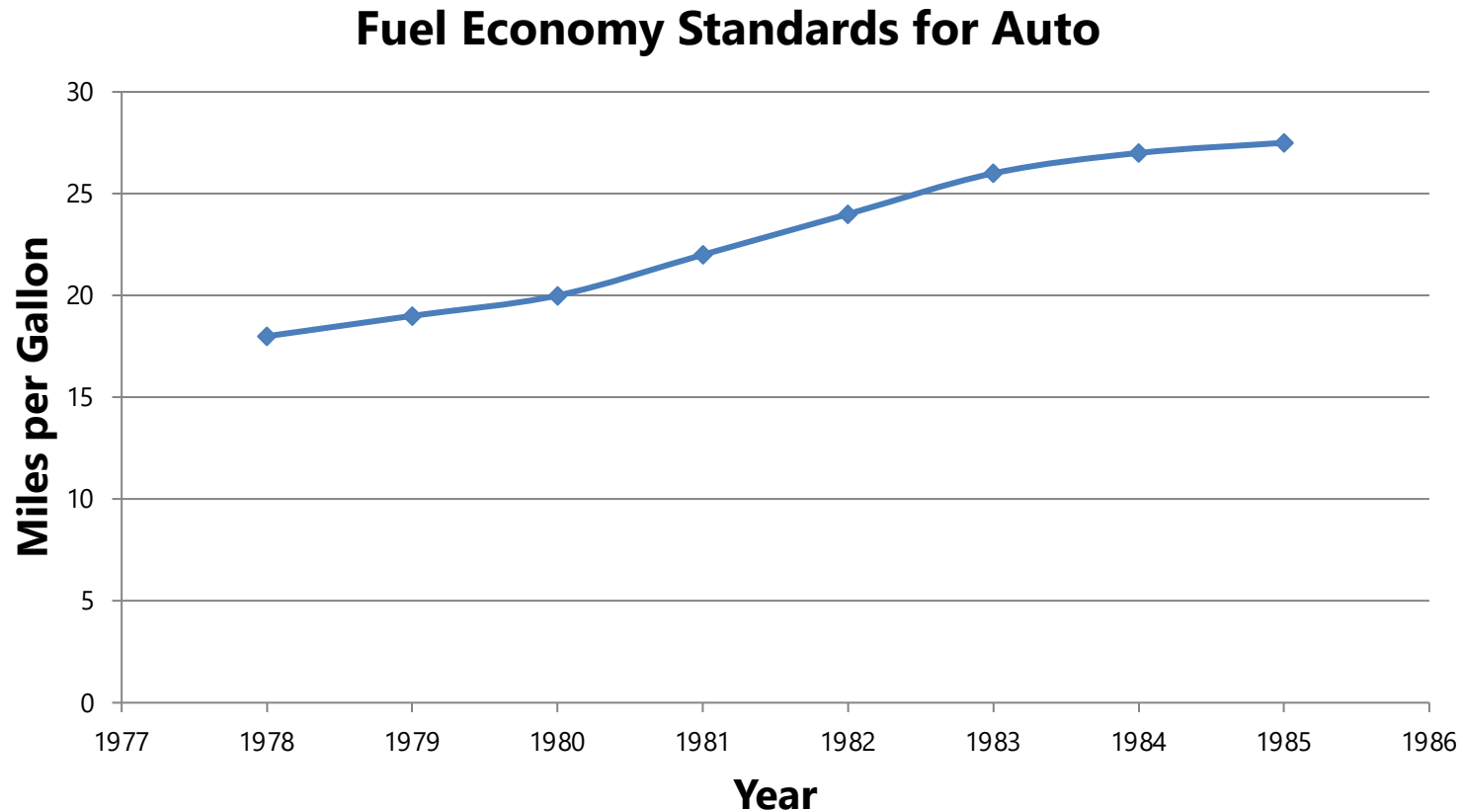
Graphic Distortion



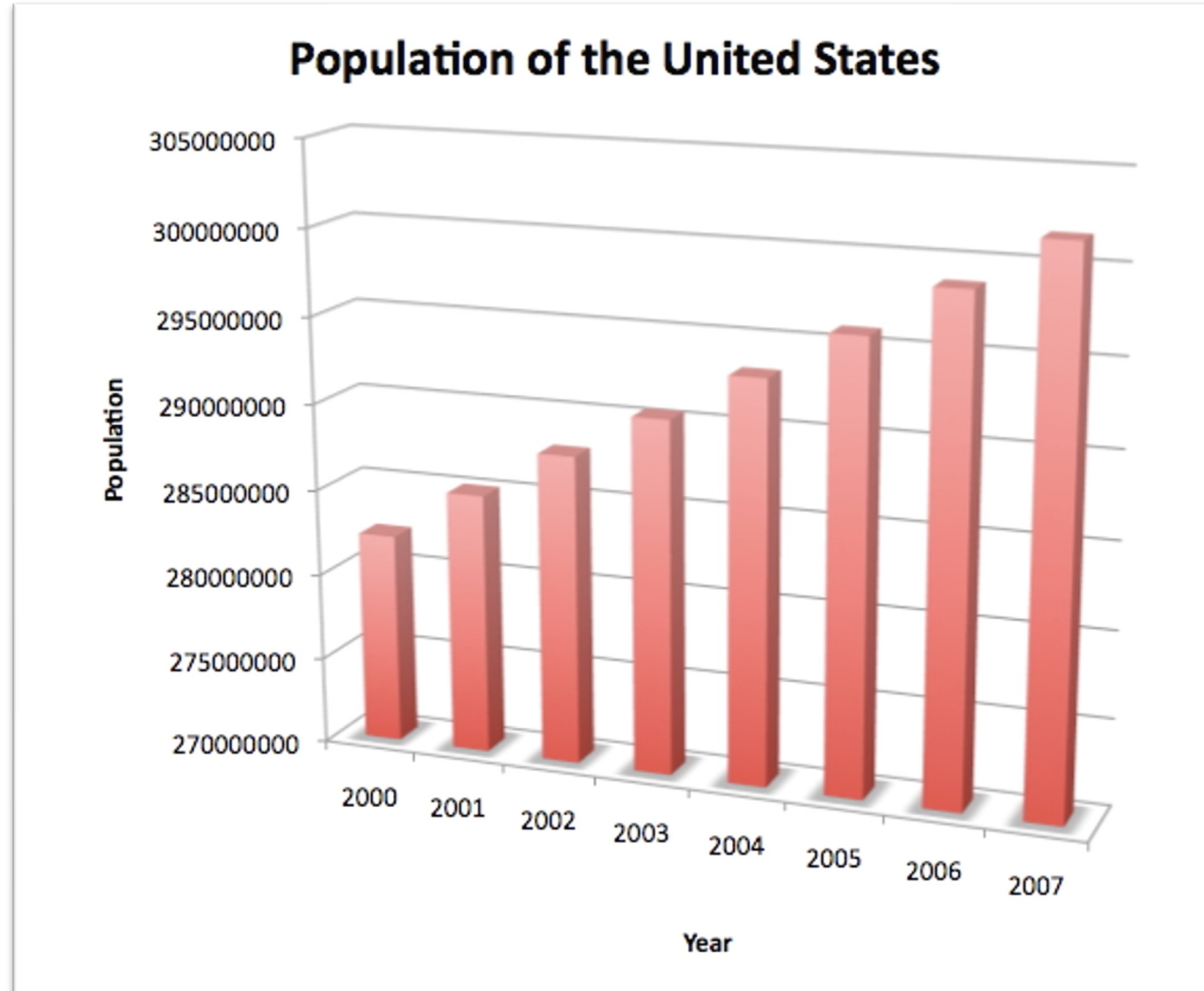
- % increase from 1978 to 1985 = $(27.5 - 18) / 18 * 100\% = 53\%$
- % increase of change in graph = $(5.3 - 0.6) / 0.6 = 783\%$
- **Lie factor** = size of effect in graphic / size of effect in data = $783 / 53 = 14.8$

Graphic Distortion

- Can be done as simple as this:



Graphic Distortion



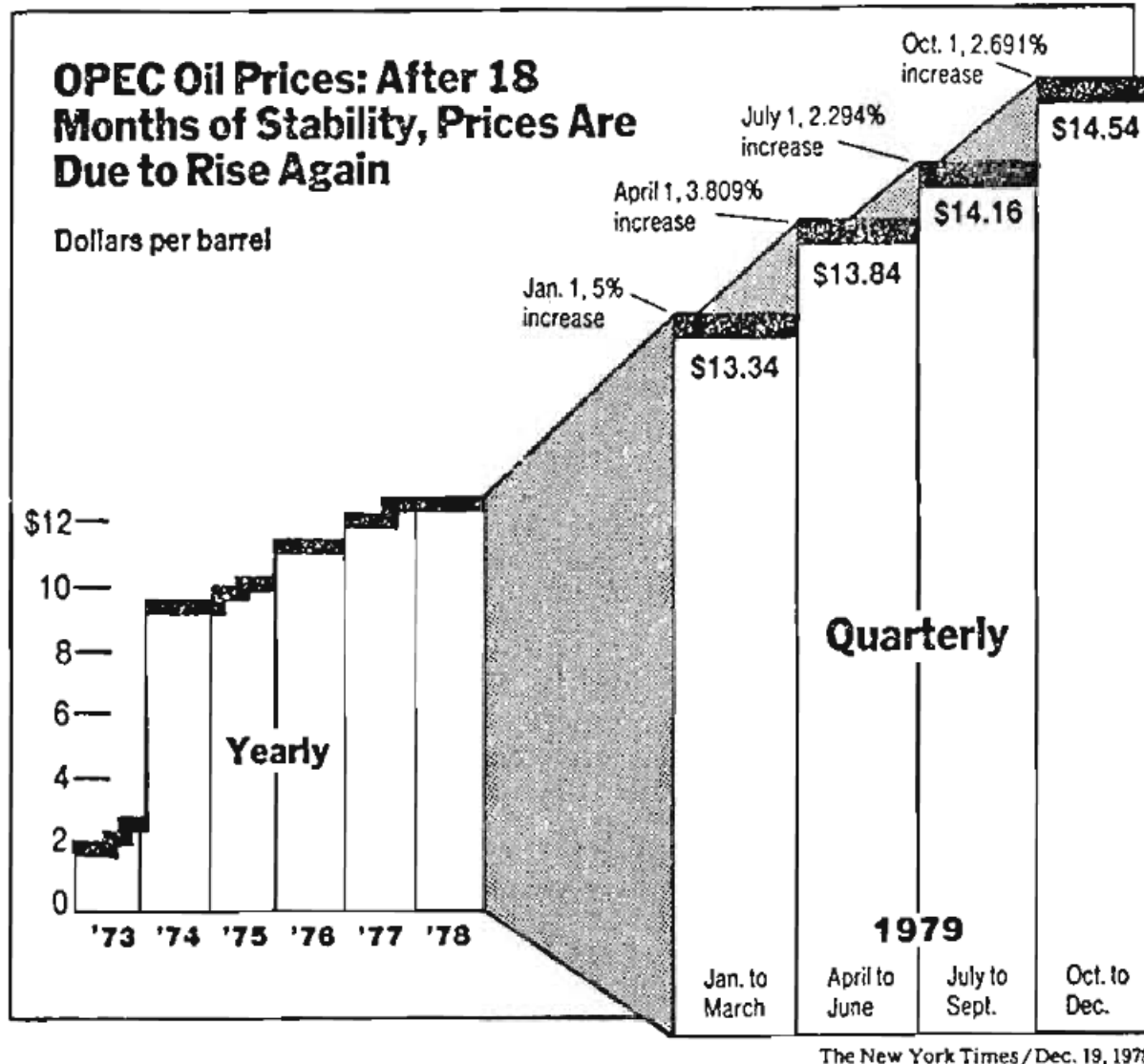
Looks like a steep change

Illusion due to 3D perspectives and the non zero-based y-axis

Design vs. Data Variations

- A design is expected to be consistent over the entire graphic.
E.g., the intervals depicted by an axis in a graph should be of uniform scale
- Should show **data variation**, but not design variation.

Design vs. Data Variations

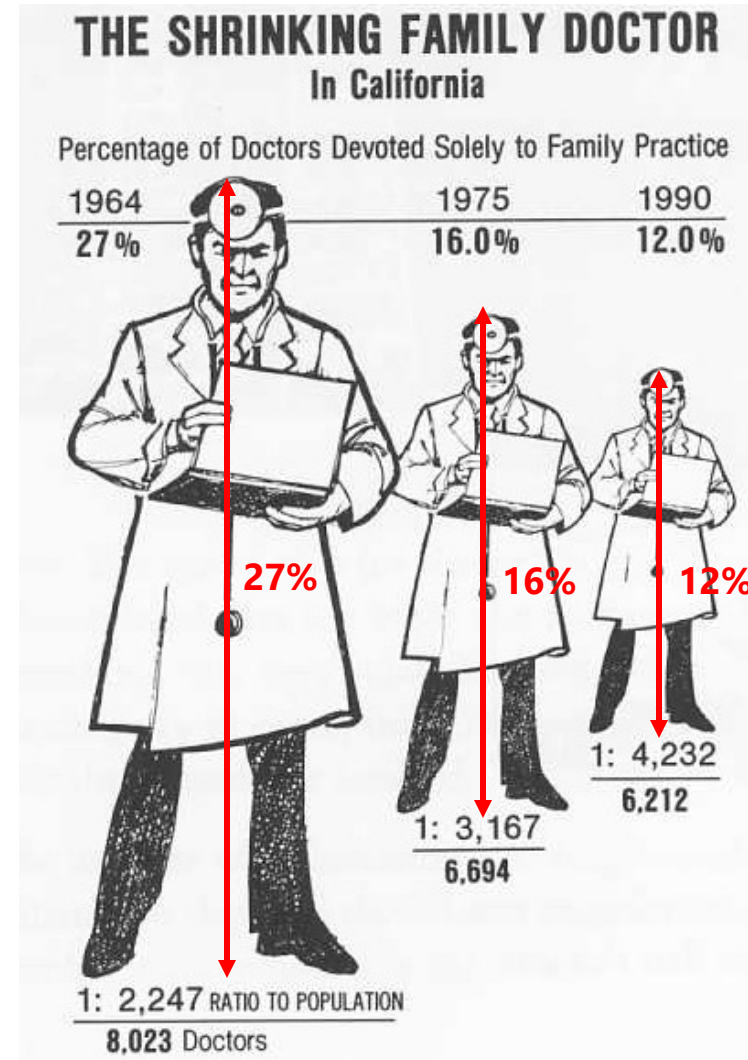


Problem:

Overloads viewers with two graphics of different horizontal scales & vertical scales in a seemingly integrated chart.

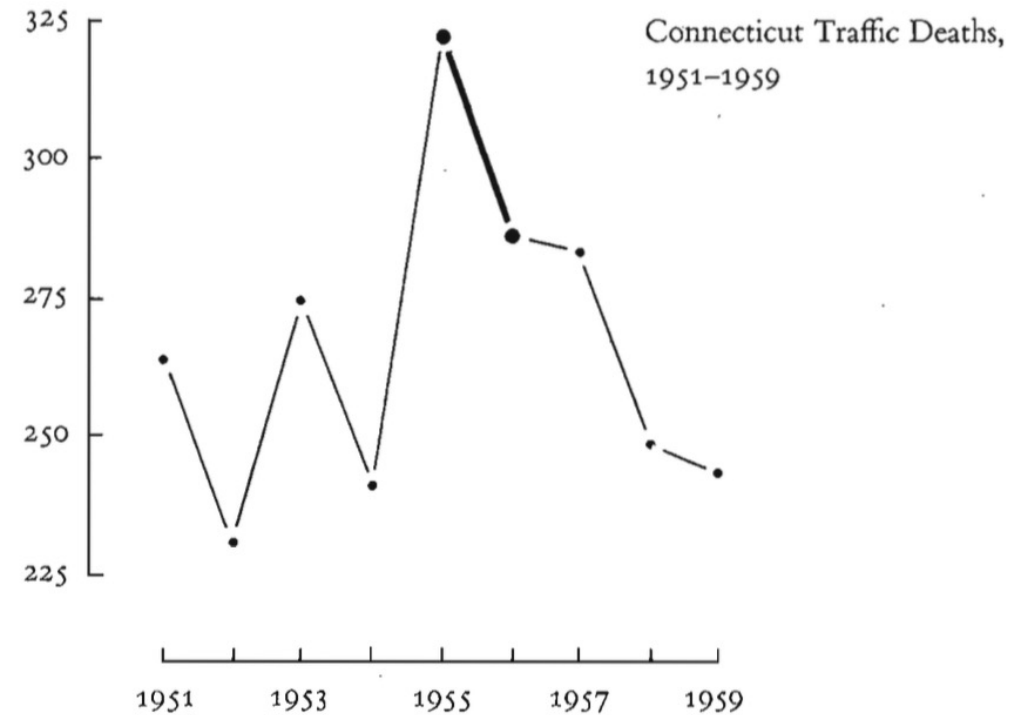
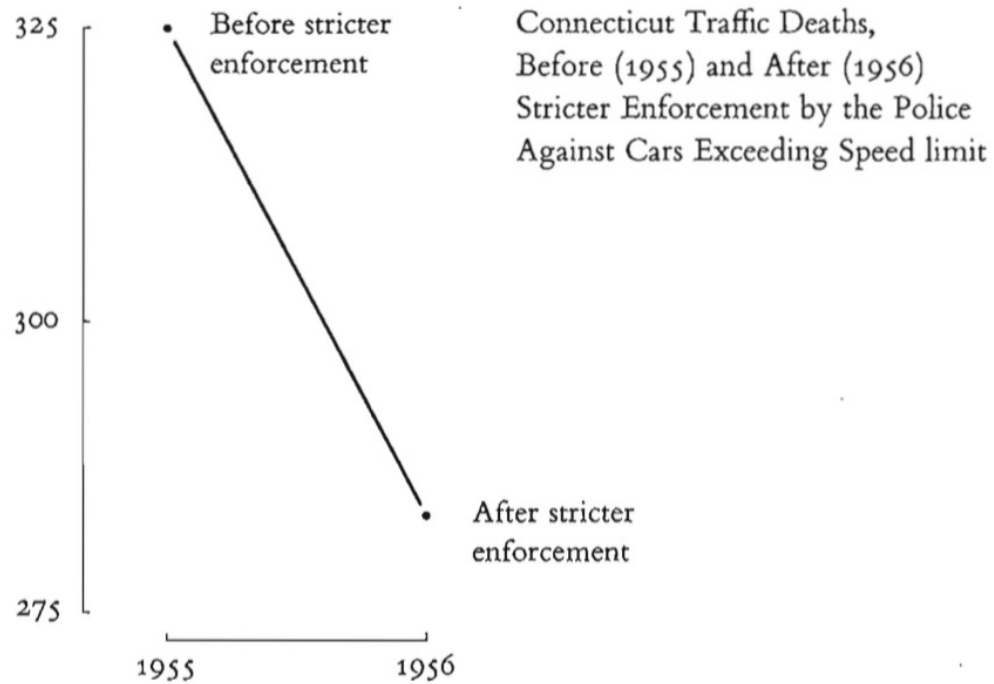
Visual Area vs. Numerical Measure

- Using areas to show one-dimensional data variation is confusing.



Context is Essential

- Ask: "Compare to what?"



Principles of Graphic Integrity

- The representation of numbers, as measured on the surface of the graphic itself, should be **directly proportional to the numerical quantities** represented.
- Clear, detailed and thorough **labeling** should be used to defeat graphical distortion and ambiguity.
- Graphics must not quote data out of **context**.

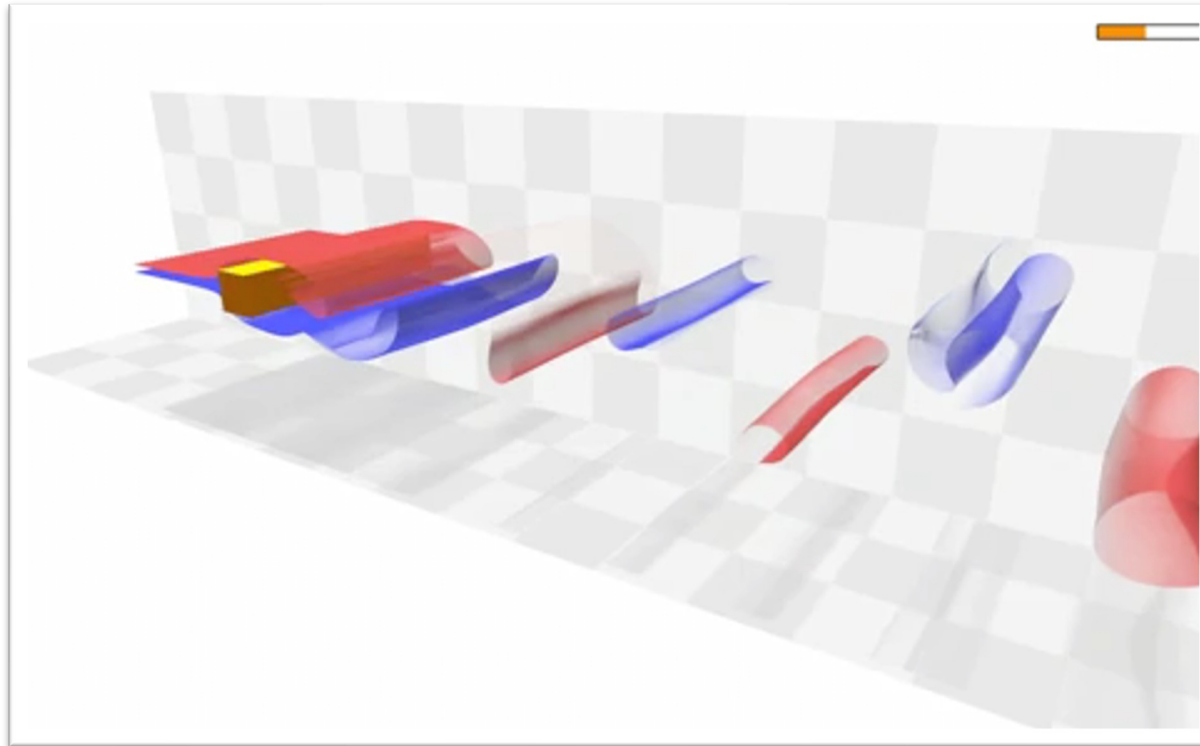
Visualization vs. Computer Graphics

- **Computer graphics**
 - Digital synthesis and manipulation of visual contents (geometry/imaging/rendering/animation)
 - Visual realism is one of the primary goals
 - Big impact in animation/movies/video games
- **Visualization**
 - Applies computer graphics techniques to generate visual display of data
 - Emphasizes on effective communication of information

Visualization vs. Computer Graphics

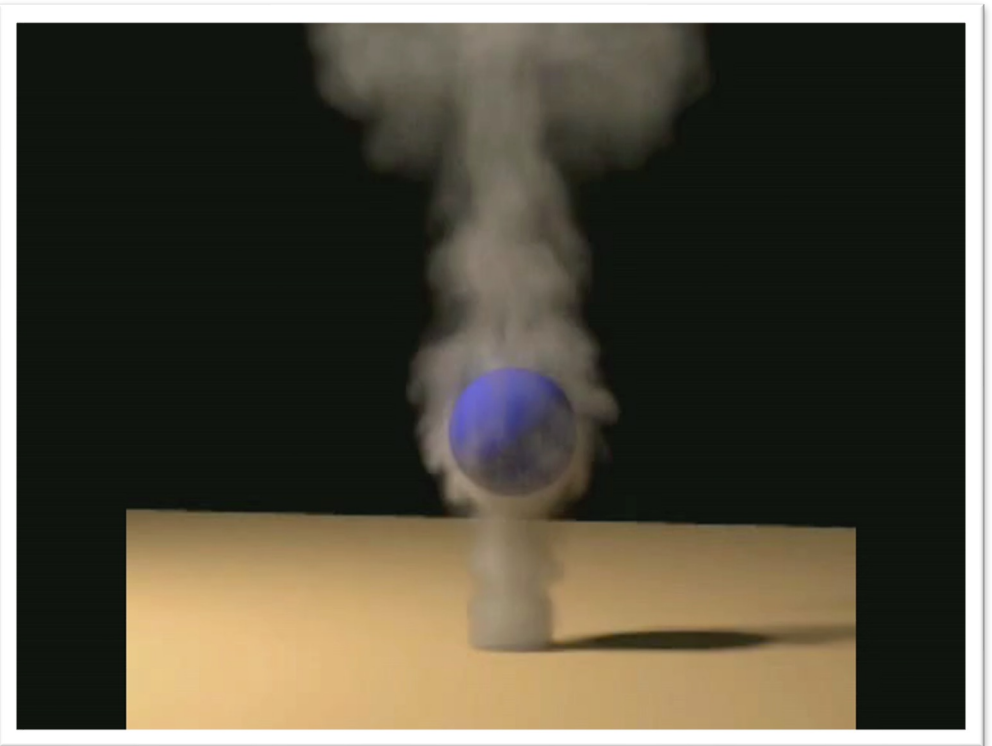
- An example:

Smoke visualization



[von Funck et al., VIS 2008]

Smoke rendering



[Fedkiw et al., SIGGRAPH 2001]

State-of-the-Art Research

- Visualization
 - IEEE VIS (InfoVis / SciVis) (ieevis.org)
- Visual Analytics
 - IEEE VIS (VAST) (ieevis.org)
 - Visual Analytics Community (vacommunity.org)
- Computer Graphics
 - SIGGRAPH (siggraph.org)

References

- Matthew Ward, Georges Grinstein and Daniel Keim, "*Interactive Data Visualization: Foundations, Techniques, and Applications*", 2010 [Chapter 1]
- Edward R. Tufte, "*The Visual Display of Quantitative Information*", 2001.
- D. Keim, J. Kohlhammer, G. Ellis and F. Mansmann, "*Mastering the Information Age: Solving Problems with Visual Analytics*", 2010.